



NSS COLLEGE OF ENGINEERING
PALAKKAD – 678008

ECL411 - ELECTROMAGNETICS LAB



LAB MANUAL

ELECTRONICS AND COMMUNICATION
DEPARTMENT

LIST OF EXPERIMENTS

EXP NO	EXPERIMENT NAME
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02.	Verification of the relation between guide wavelength, free space wavelength and cut-off wavelength for rectangular waveguide
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MICROWAVE EXPERIMENTS

EXPERIMENT – 1

REFLEX KLYSTRON MODE CHARACTERISTICS

AIM

To construct a microwave bench set up using Reflex Klystron as source and perform the mode characteristics.

EQUIPMENTS REQUIRED

1. Klystron Power Supply
2. Klystron with mount
3. Isolator
4. Frequency meter
5. Variable Attenuator
6. Slotted section with Probe carriage
7. CRO
8. Matched Termination

THEORY

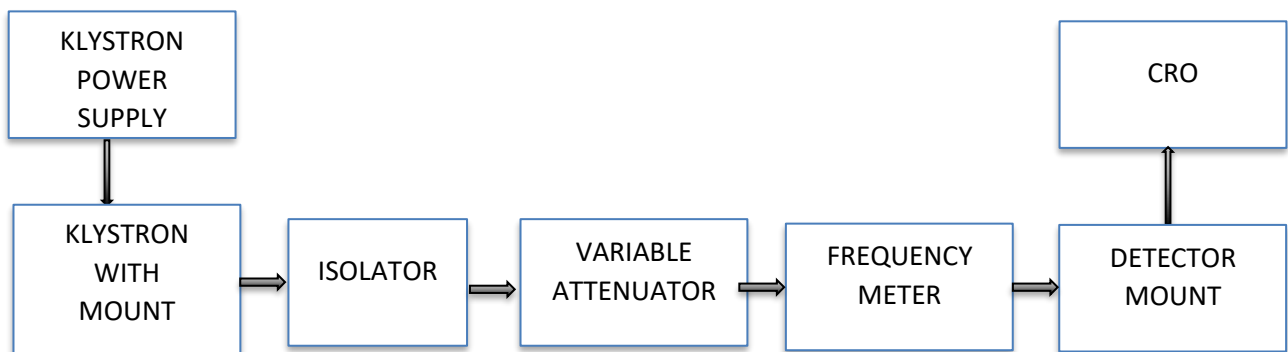
Klystron is a microwave vacuum tube employing velocity modulation. These electrons move towards the repeller (ie) the electrons leaving the cavity during the positive half cycle are accelerated while those during negative half cycle are decelerated. The faster ones penetrate further while slower ones penetrate lesser in the field of repeller voltage. But, electrons leaving faster from the cavity take longer time to return and hence catch up with slower ones. In the cavity the electrons bunch and interact with the voltage between the cavity grids. It consists of an electron gun producing a collimated electron beam.

It bunches pass through grids at time the grid potential such that electrons are decelerated and they give back energy. The electrons are then collected by positive cavity wall near cathode. To protect repeller from damage, repeller voltage is applied before the accelerating voltage.

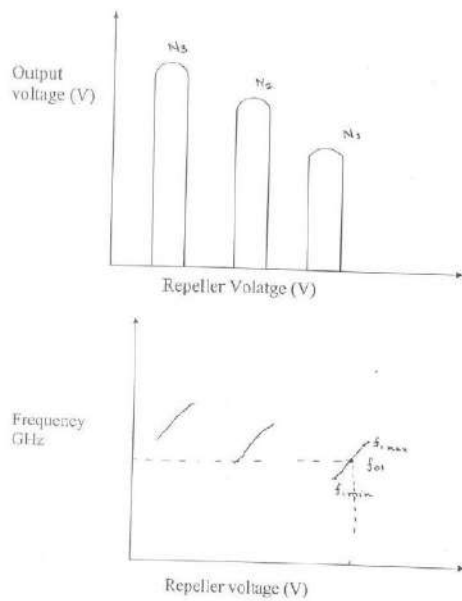
PROCEDURE

1. Assemble components as in the figure.
2. After necessary precautions, switch ON the power supply.
3. To obtain peak voltage, attenuator positioned at minimum attenuation.
4. Vary the repeller voltage from maximum negative and increase steps on N and record output power and frequency.
5. Frequency measured by tuning basic frequency meter to have dip in output voltage each time.
6. Frequency meter is defined before measuring output power each time.
7. Mode characteristics of Reflex Klystron is plotted.

BLOCK DIAGRAM



MODEL GRAPH



OBSERVATIONS

Mode	Repeller voltage(V)	Output voltage(mV)	Frequency (GHz)	Power(mW)
1				
2				
3				

RESULT

The mode characteristics of Reflex Klystron are plotted in graph.

EXPERIMENT – 2

VERIFICATION OF THE RELATION BETWEEN GUIDE WAVE LENGTH, FREE SPACE WAVE LENGTH AND CUT OFF WAVE LENGTH

AIM

To verify the relation between Guide wave length, free space wave length and cut off wave length for rectangular wave guide.

EQUIPMENTS REQUIRED

1. Klystron Power Supply
2. Klystron with mount
3. Isolator
4. Frequency meter
5. Variable Attenuator
6. Slotted section with Probe carriage
7. CRO

8. Matched Termination

THEORY

The following relationships are important for rectangular waveguide.

$$C = f\lambda$$

$$\frac{1}{\lambda_o^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$$

$$\lambda_g = \frac{\lambda_o}{\sqrt{1 - (\lambda_o/\lambda_c)^2}}$$

$$\lambda_c = \frac{2}{\sqrt{(m/a)^2 + (n/b)^2}}$$

where C = velocity of light

f = frequency

λ_o = wavelength in free space

λ_g = wavelength in wave guide

λ_c = cut-off wavelength in waveguide

a = broader dimension in the cross-section of waveguide

b = smaller dimension in the cross-section of waveguide

For TE_{10} mode, by substituting the values of m and n , we get

$$\lambda_c = \frac{2}{\sqrt{(1/a)^2 + (0/b)^2}} = 2a$$

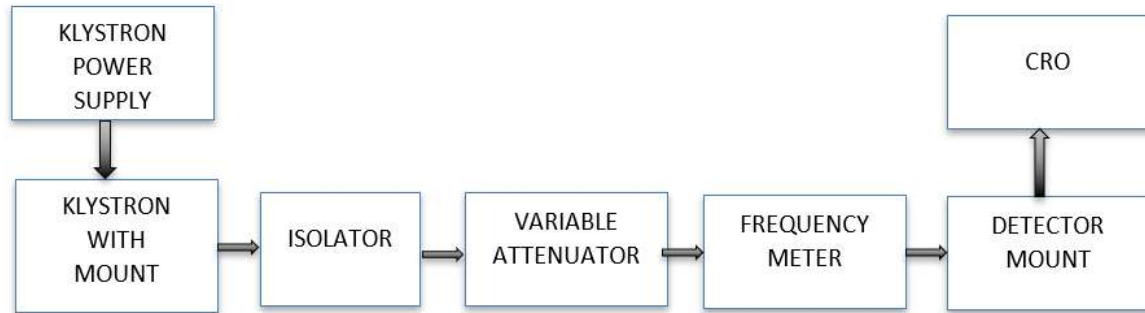
$$\lambda_g = \frac{\lambda_o}{\sqrt{1 - (1/2a)^2}}$$

$$\lambda_o = \frac{1}{\sqrt{(1/\lambda_g)^2 + (1/2a)^2}}$$

$$f = C/\lambda_o = C \sqrt{(1/\lambda_g)^2 + (1/2a)^2}$$

The guide wavelength can be measured as twice the distance between two successive maxima (or) minima in the standing wave pattern.

BLOCK DIAGRAM



PROCEDURE

1. Switch ON the power supply.
2. Obtain square wave in the CRO.
3. Maximize the output by changing the Klystron bias, attenuator, tunable probe detector plunger, tunable probe penetration.
4. Measure frequency by observing zero or close to zero-amp output in CRO.
5. Replace the termination with movable short and detune the frequency meter.
6. Move the probe along with the slotted section. The deflection in CRO will vary.
7. Move the probe position to maximum and again move probe to next maximum position and note MSR & VSR.
8. Calculate the guide wavelength as twice the direction between two maximum positions.
9. Measure the waveguide inner broad dimension 'a'.
10. Calculate the frequency f_0 .

OBSERVATIONS AND CALCULATIONS

$$f_0 = \underline{\hspace{2cm}}$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$\lambda_0 = \frac{C}{f_0} = \underline{\hspace{2cm}}$$

Main Scale Reading	Vernier Scale Reading	Total Reading
		$d_1 = \underline{\hspace{2cm}}$
		$d_2 = \underline{\hspace{2cm}}$
		$d_3 = \underline{\hspace{2cm}}$

$$\lambda_g = (d_1 - d_2) + (d_2 - d_3)$$

$$= \underline{\hspace{2cm}}$$

$$\frac{1}{\lambda_c^2} = \frac{1}{\lambda_0^2} - \frac{1}{\lambda_g^2}$$

$$\text{Therefore, } \lambda_c = \underline{\hspace{2cm}}$$

$$f_c = \frac{C}{\lambda_c} = \underline{\hspace{2cm}} \text{ GHz}$$

THEORETICAL CALCULATION

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

For TE_{10} mode, $m=1$, $n=0$

$$\text{Hence, } \lambda_c = 2a$$

$$a = \underline{\hspace{2cm}}$$

$$\lambda_c = \underline{\hspace{2cm}}$$

$$f_c = \frac{C}{\lambda_c} = \underline{\hspace{2cm}} \text{ GHz (theoretically)}$$

RESULT

Verified the relationship between the guide wavelength(λ_g), free space wavelength(λ_0) and cut off wavelength(λ_c) for a rectangular waveguide.

EXPERIMENT 3

VSWR AND FREQUENCY MEASUREMENT

AIM

To measure the frequency and the standing wave ratio in a microwave transmission line.

EQUIPMENTS REQUIRED

Microwave source (Klystron Power Supply) , Klystron Mount, Isolator, Variable attenuator, slotted section, matched termination, VSWR meter or CRO.

THEORY

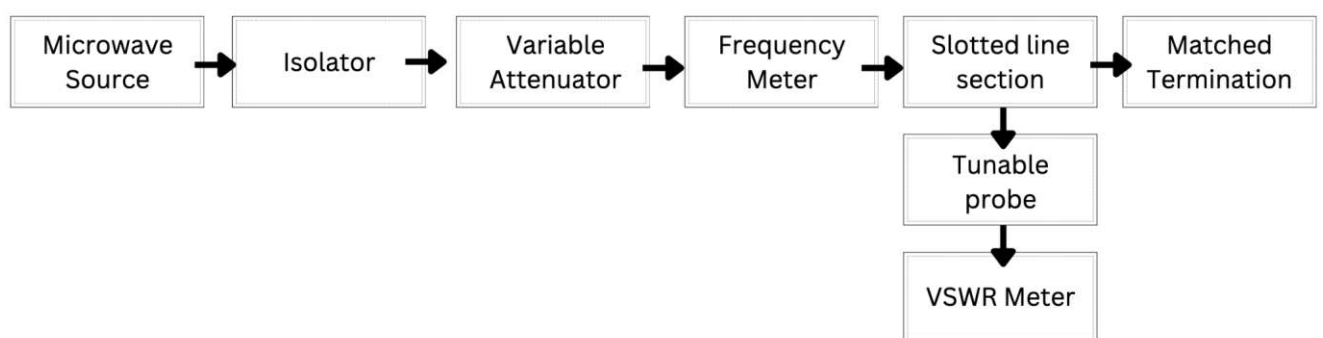
The electromagnetic field at any point of the transmission line may be considered as the sum of two traveling waves: the incident wave which propagates from generator and the reflected wave which propagates towards the generator. The reflected wave is set up by the reflection of the incident wave from a discontinuity on the line or from the load impedance. The magnitude and phase of the reflected wave depends upon the amplitude and phase of the reflected impedance. The presence of two traveling waves give rise to the standing waves along the line. The maximum field strength is found when two waves are in phase and the minimum where the two waves add in the opposite phase. The voltage standing wave ratio is the ratio between the maximum and minimum field strength along the line.

$$VSWR = V_{max} / V_{min}$$

Frequency measurement

Frequency meter is made up of a cylindrical cavity [absorption type]. By varying the effective height (d) of the cavity, its resonance frequency may be varied. At resonance, i.e., when incoming signal frequency is matched with resonance frequency of the cavity, maximum energy is stored in the cavity resulting in a minimum transmitted output.

BLOCK DIAGRAM



PROCEDURE

VSWR Measurement

1. Set up the components and equipments as shown in figure.
2. Keep the variable attenuator in the minimum attenuation position.
3. Keep the control knobs of VSWR meter as below:
 - Range (dB) - 40dB/ 50 dB
 - Input switch -Low impedance
 - Meter switch -Normal
 - Gain(Coarse- fine) -Mid position approx.
4. Keep the control knobs of Klystron Power Supply as below:
 - Beam voltage -OFF
 - Mod- switch -AM
 - Beam voltage Knob - Fully anticlockwise
 - Reflector voltage Knob- fully clockwise
 - AM - Amplitude Knob -Around fully clockwise
 - AM - Frequency Knob - Mid position.
5. Switch on the Klystron power supply, VSWR meter.
6. switch on the beam voltage switch position and set the beam voltage at 300V.
7. Rotate the reflector voltage knob lo get deflection in VSWR meter.
8. Tune the output by turning the replaces voltage knob, amplitude and Frequency of AM modulation.
9. Tune the plunger of Klystron Mount and probe for maximum deflection in VSWR meter.
10. If required, change the range dB - Switch variable attenuator position and gen control knob to get the maximum deflection in scale of VSWR meter
11. Move the probe along the slotted line to get maximum deflection in VSWR meter.
12. Adjust the vswR meter gain control knob or variable attenuator until the meter indicales 1.0 on normal VSWR on scale.

Frequency Measurement

1. Set up the components as shown in figure.
2. Set the variable attenuates at minimum attenuation position.
3. Keep the control knob of VSWR as below:
 - Range -50 dB
 - Input switch -Crystal low impedance
 - Meter switch -Normal position
 - Gain (coarse fine) -Mid position
4. Keep the control knobs klystron power supply as below:
 - Beam Voltage -OFF
 - Meter Switch -AM
 - Beam voltage knob -Fully anticlockwise
 - Reflector voltage - Around mid-position.
5. Switch ON the Klystron Power supply and CRO
6. Switch ON the beam voltage switch and set beam voltage at 300v with the help of beam voltage knob.

7. Adjust the voltage to get square waveform in CRO.
8. Tune the frequency meter knob to get a dip on the CRO and note down the frequency directly from frequency meter.

RESULT

The voltage standing wave ratio and frequency for the given matched load is found.

EXPERIMENT 4

DIRECTIONAL COUPLER CHARACTERISTICS

AIM

To construct a microwave bench setup using a Multi hole directional coupler and measure the performance parameters:-

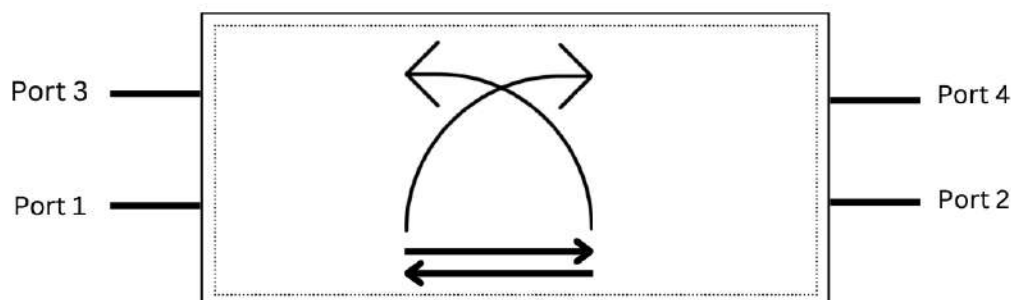
1. Directivity
2. Coupling Factor
3. Isolation

EQUIPMENTS

Klystron power supply, Klystron Tube, Klystron Mount, Isolator, Frequency meter, Variable attenuator, Slotted section, Detector mount, Waveguide Stands, VSWR meter, BNC Cable, CRO, Directional coupler.

THEORY

It is a passive four port device. Directional coupler is a reciprocal device. It consists of a primary guide with port 1 and 2 and a secondary guide with port 3 and 4. It is made of two connected waveguides; one of the waveguides curves away. The waveguides are coupled through holes between them. The directional coupler is said to be consisting of main arm and auxiliary arm as shown



The amount of power coupled to the auxiliary arm depends on the number of holes and their size. The matched termination absorbs power without reflection. The coupler is used to find out the power reflection coefficient in a waveguide and find out incident and reflected power value. The power at port 4 and port 2 have phase difference of 90 degree. Similarly, the power at port 3 and 1 have a phase difference of 90 degree when propagation is in reverse direction. The guides 1-2 and 3-4 are identical. Any one of them can be used as primary and other acts as auxiliary guide.

The parameters of directional coupler:

1. Coupling Factor (C):

$$C = 10 \log_{10} \frac{P_i}{P_f} \text{ dB}$$

2. Directivity (D):

$$D = 10 \log_{10} \frac{P_f}{P_b} \text{ dB}$$

3. Isolation (I):

$$I = 10 \log_{10} \frac{P_i}{P_b} \text{ dB}$$

$$I \text{ (in dB)} = C \text{ (in dB)} + D \text{ (in dB)}$$

P_i = Incident power at Port 1

P_r = Received power at Port 2

P_f = Forward coupled power at Port 4

P_b = Back power at Port 3

PROCEDURE

1. Connect the components as shown in figure.
2. Keep the control knobs klystron power supply as below:
 - Modulation Selection -AM
 - Beam voltage knob -Fully anticlockwise
 - Reflector voltage knob -Fully clockwise
 - Selector switch-Beam Voltage
3. Keep the AM modulation control knobs of the amplitude and frequency at mid-position
4. Switch on the Klystron power supply.
5. Now the beam voltage knob is set to 295V or 200V and a reflection voltage of 150V. Then change the selected knob to beam current (the beam current should not be more than 30 mA).
6. Now observe the square waveform in CRO by varying either reflector voltage or adjusting the amplitude of AM.
7. Connect the slotted section output to detector mount, now measure the input power P_i .
8. Disconnect to set up as shown in figure.
9. Now connect the slotted section output to directional coupler input, connect the detector mount at Port 2, terminate the port 4 with matched termination.
10. Now measure the power at Port 2, then remove detector Mount from port 2 and connected to port 4. Terminate port 2 with match determination.
11. Now measure power at Port 3.
12. Now remove the directional coupler from slotted section and connect the slotted section output to directional coupler port 2, connect the detector mount at port 1, terminate port 4 with a matched termination.
13. Now measure the power at Port 4 and tabulate the recordings.

TABULATION

Beam voltage =

Beam current =

Repeller voltage =

Sl. No	Parameters to be measured	Voltage	Power
1	Back power		
2	Coupled power		
3	Incident power		

RESULT

The circuit was established and the following directional coupler characteristics were measured.

Directivity =

Isolation Factor =

Coupling Factor =

OPTICAL EXPERIMENTS

EXPERIMENT NO 1 SETTING UP OF FIBER OPTIC DIGITAL LINK

AIM

To study a 660nm fiber optic digital link and to study the relationship between the input signal and received signal.

COMPONENTS REQUIRED

1. Kit
2. Dual channel oscilloscope
3. Fiber cable
4. Power supply

THEORY

Fiber optic link can be used for transmission of digital as well as optical signals. Basically, a fiber optic link contains three main elements, a transmitter, an optical fiber and a receiver. The transmitter module takes the input signals in electrical form and then transform it into optical energy containing the same information. The optical fiber is the medium, which takes the energy to the receiver and the receiver light is converted back into electrical form with the same pattern as originally fed to the transmitter.

TRANSMITTER:

Fiber optic transmitters are typically composed of a buffer, driver and optical sources. The buffer provides both an electrical connection and isolation between the transmitters and the electrical system supplying the data. The driver provides electrical power to the optical source. Finally, the optical source converts the electrical current to the light energy with the same pattern. Commonly used optical source are light emitting diodes (LEDs) and laser beam. Simple LED circuits for digital and analog transmissions are shown in the figure. It shows trans conductance drive circuits for analog transmission-common emitter configuration. The transmission section comprises of:

1. Function generator
2. Frequency modulator
3. Pulse width modulator block

The function generator generates the input signals that are going to be used as information to transmit through the optic link. The output voltage available is 1KHz sinusoidal signal of adjustable amplitude, and fixed amplitude 1KHz square wave signal. The modulator section accepts the information signals and convert it into suitable form for transmission through the fibre optic link.

THE FIBRE OPTIC LINK:

FO transmitter and FO receiver circuit on board form the fibre optic link. This section provides the light source for the optic fibre and the light detector at the far end of the fibre optic links. The optic fibre plugs into the connectors provided in this part of the board. Two separate links are provided.

RECIEVER:

The comparator circuit, low pass filter, phase locked loop, amplifier circuits from receiver on the board. It is able to undo the modulation process in order to recover the original information signal. In this experiment the trainer board is used to illustrate one-way communication between digital transmitter and receiver circuits.

EXPERIMENTAL SETUP

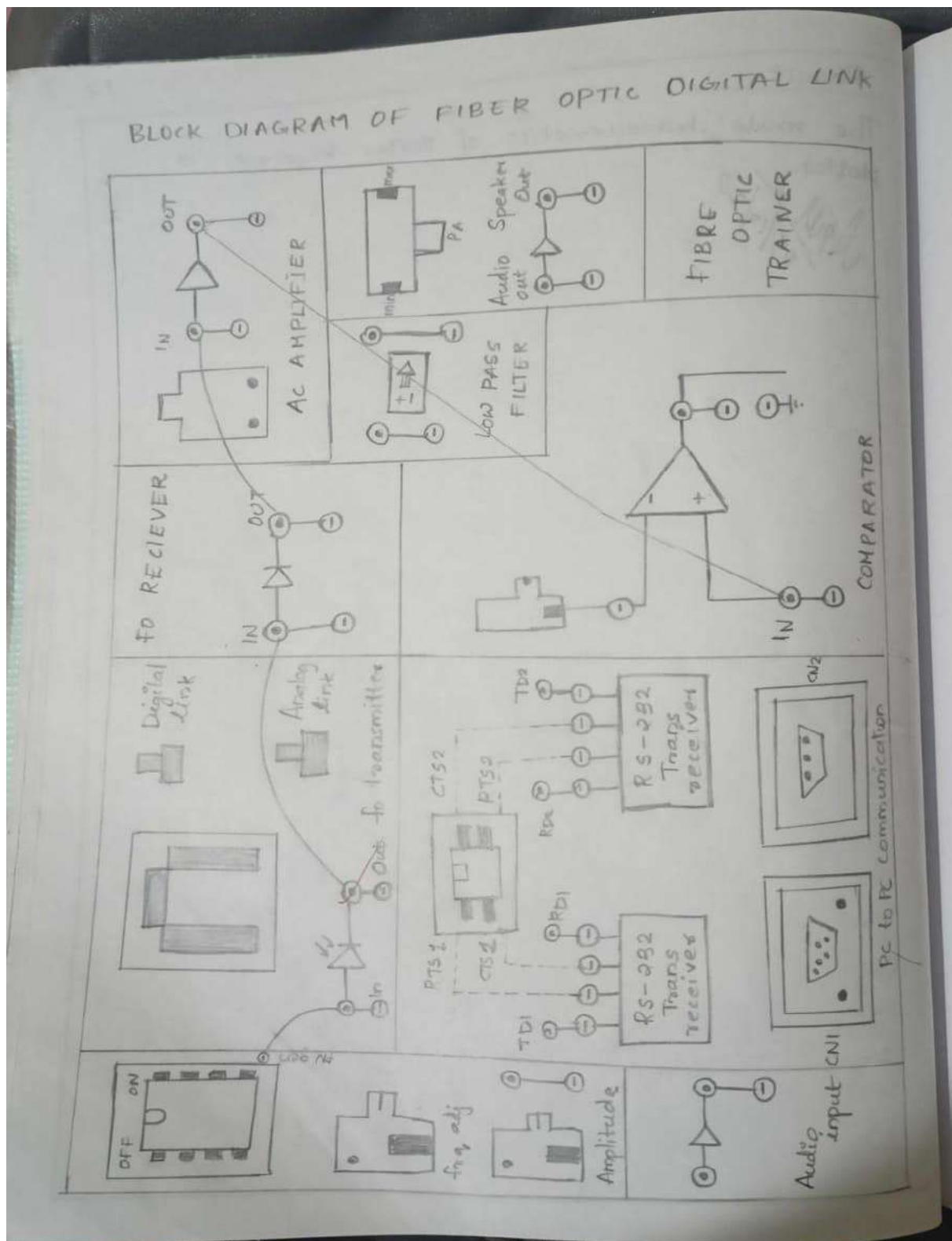


Figure 1.1 Experimental Setup of Fiber optic digital link

PROCEDURE

1. Connects the AC supply to kit.
2. Ensure that all fault switch is in normal position.
3. Make the connects as shown in block diagram.

4. Connect the 1KHz square wave output source to 660nm fiber optic transmitter input.
5. Connect 1meter, fiber optic cable between fiber optic transmitter output and fiber optic receiver input
6. Connect fiber optic receiver output to AC amplifier input.
7. Connect AC amplifier output to comparator input.
8. Set the switch SW2 to digital mode.
9. Switch on the power.
10. Observe the signal at comparator output on CRO, slowly adjust the comparator pot until waveform matches the input source.
11. Vary the input frequency and observe the output.

OBSERVATION TABLE

INPUT FREQUENCY	INPUT SIGNAL (V _{IN})	TRANSMITTED SIGNAL (V _{TRA})	RECEIVED SIGNAL (V _{REC})	AMPLIFIED SIGNAL (V _{AMP})	COMPARATOR OUTPUT (V _{COMP})

RESULT

Thus, the 660nm fiber optic digital link was studied and observed and plotted the relationship between the input signal and output signal.

EXPERIMENT NO 2

MEASUREMENT OF NUMERICAL APERTURE OF A

FIBER OPTIC CABLE

AIM

To measure the numerical aperture of the plastic fiber provided with the kit using 600nm wave length LED.

COMPONENTS REQUIRED

1. Kit
2. 1-meter fiber cable
3. NA J1G
4. Steel Ruler
5. Power Supply

THEORY

Numerical aperture refers to the maximum angle at which light incident on the fiber is totally internally reflected and is properly transmitted along the fiber. The cone formed by the rotation of this angle along the axis of the fiber is the cone acceptance of the fiber. The light ray should strike the fiber within its cone of acceptance, else it is refracted out of the fiber core.

CONSIDERATION IN NA MEASUREMENT

It is very important that optical source should be properly aligned with cable and distance from the launched point and cable is properly selected to ensure that the maximum amount of optical fiber power is transferred to the cable.

The experiment is best performed in less illuminated room.

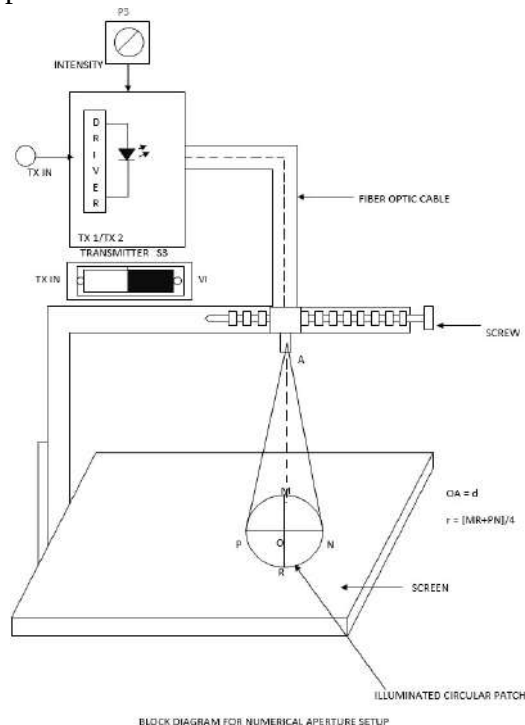


Figure: Block diagram of numerical aperture set-up

PROCEDURE

1. Slightly unscrew the cap of LED SFH 756V(600nm). Do not remove the cap from the connector. Once the cap is loosened insert the fiber into the cap. Now light the cap by screwing it back.
2. Connect the power supply cable with proper polarity to kit while connecting this ensure that power supply is OFF.
3. Keep pot p3 fully clockwise position and p4 fully anticlockwise position.
4. Switch on the power supply
5. Insert the other end of the fiber into numerical aperture measurement. Hold while sheet facing the fiber. Adjust fiber such that its cut fall is perpendicular to axis of fiber.
6. Keep distance of about 10mm between fiber tip and screen. Gently tight the screw and thus fix fiber in place.
7. Now adjust pot p4 fully clockwise position and observe the illuminated path of light on screen.
8. Measure exactly distance d and also vertical and horizontal diameter MR and PN.
9. Mean radius is calculated using the formula

$$r = (MR + PN) / 4$$

10. Find the numerical aperture of fiber using the formula

$$NA = \sin \theta_{\max} = \frac{r}{\sqrt{(d^2 + r^2)}}$$

11. $\theta_{\max}$ is the maximum angle at which light incident is properly transmitted through fibre.

OBSERVATION TABLE

d (mm)	MR (mm)	PN (mm)	NA	$\theta_{\max}$

Mean value of NA =

Mean value of $\theta_{\max}$ =

RESULT

The mean value of numerical aperture of given optical fibre is and $\theta_{\max}$ =

EXPERIMENT NO 3

VOLTAGE VS. CURRENT (V-I) CHARACTERISTICS OF FIBER OPTIC PHOTO DIODE

AIM

To plot the characteristics of pin photodiode

COMPONENTS REQUIRED

1. Optical fiber
2. Multimeter
3. Kit
4. Power supply

THEORY

A PIN diode operates under high level injection. The intrinsic level injection region is flooded with charge carrier from the P and N regions. When the diode is forward biased, the injected carrier concentration is several orders of magnitude higher than the intrinsic carrier concentration. Due to this high-level injection, which in turn is due to the depletion process, the electric field extends deeply into the region. These electric fields help in speeding up of the transport of charge carriers from the P and N region, which results in faster operation of the diode, making it suitable for high frequency operation. The PIN diode obeys the standard diode equation for low frequency signals. At higher frequencies, the diode looks like an almost perfect resistor. The PIN diode has a relatively large stored charge drift in an intrinsic region. At low frequency, the stored charge can be fully swept and the diode turns off. At higher frequencies, there is not enough time to sweep the charge from the drift region, so the diode never turns off. The time required to sweep the stored charge from a diode junction is its reverse recovery time.

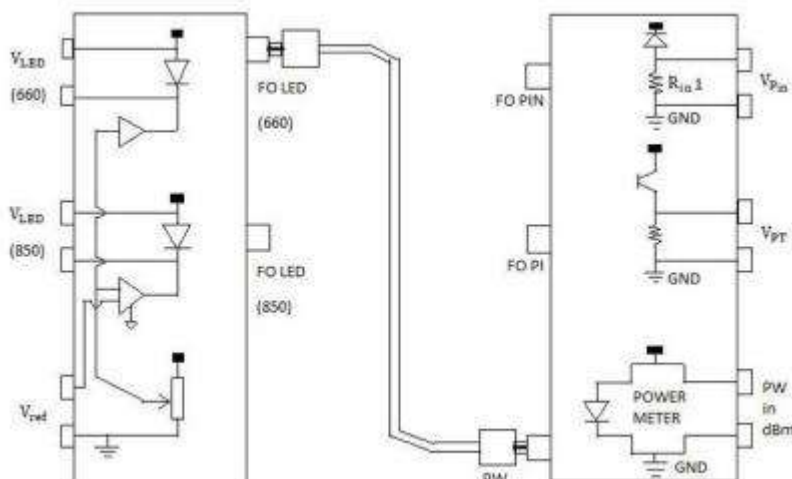


Figure: Schematic of experimental set-up

OBSERVATION TABLE

V_{ref} (mV)	$I_f = V_{ref}/100$ (mA)	V_{pin}

PROCEDURE

1. Power the system up and connect a DMM across V_{ref} in the range 2000 mv.
2. Adjust and setup the I_f knob and set V_{ref} to 100 mV. This corresponds to an I_f at 1mA current in the PIN photodiode.
3. Connect one end of fiber optic cable to the fiber optic LED of 660 part and other to FO pin.
4. Record the V_f for PIN photodiode(660 nm)

RESULT

The V-I characteristics of PIN photodiode was plotted.

EXPERIMENT NO 4

V-I CHARACTERISTICS OF FIBRE OPTIC LED

AIM

To plot the V-I characteristics of fibre optic led at 660nm and 850nm

COMPONENTS REQUIRED

1. Fibre optic communication trainer kit
2. 1-meter fibre optic cable

THEORY

LED is a semiconductor diode made by creation of a junction of n-type and p-type materials. Thus, the principle of an LED's actions works the same way as that of normal diodes. That is when external energy is provided, the electrons in the conduction band get excited from there they fall to valence band and recombine with holes. The only difference is that in a normal diode the recombination release thermal energy rather than visible spectrum. Optical fibre is used as the transmission medium in fibre optic communication system. It is transparent flexible filament that guides optical information signal from transmitted side to receiver side through the principle of total internal reflection.

OBSERVATION TABLE

SL NO	660nm			850nm		
	V_{ref} (mv)	$I_f = \frac{V_{ref}}{100}$ (mA)	V_f (V)	V_{ref} (mv)	$I_f = \frac{V_{ref}}{100}$ (mA)	V_f (V)

PROCEDURE

1. Power the system up. Connect a DMM across V_{ref} in the range 2000mv.
2. Adjust and setup I_f knob and set V_{ref} to 100mv. This corresponds to an I_f at 1 mA in the fiber optic LED.
3. Connect one end to the fiber optic cable to the fibre optic LED of 660part.
4. Record the V_f for fiber optic LED (660nm).
5. Repeat the above steps for fiber optic LED 850nm.

RESULT

The V-I characteristics of fiber optic LED was plotted.

SIMULATION EXPERIMENTS

EXPERIMENT NO 1

FAMILIARISATION OF CST STUDIO SUITE

Aim

To familiarise with CST Studio Suite for the simulation of antenna experiments.

Tools used

CST Studio Suite 2022 (Learning Edition).

Procedure

CST Studio Suite is a powerful simulation platform for all kinds of electromagnetic field problems and related applications. The program provides a user-friendly interface to handle multiple projects and views at the same time.

One of the outstanding features of the environment is the seamless integration of various simulation methods and strong interoperability management.

Installing the software

The steps to install the software in Windows OS is provided below:

1. Check system compatibility
2. Browse the internet and download the CST Studio Suite installer (Note: Learning Edition is free of cost)
3. Install by following the instructions.

Creating a new project

For creating a new project, see **Fig. 1.1**

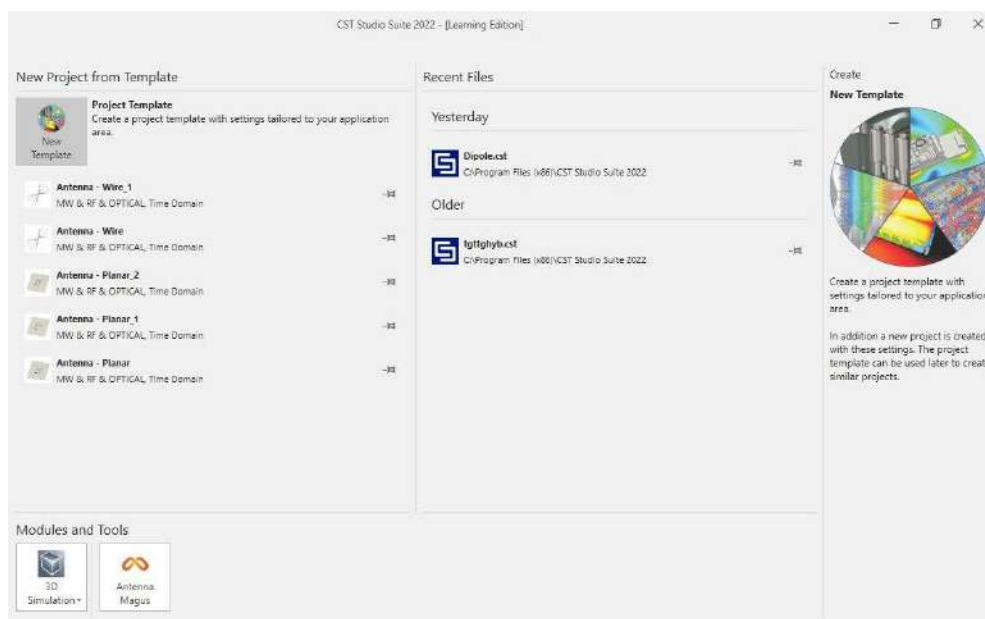


Fig. 1.1: Creating a new project

Choose a project template as shown in **Fig. 1.2** and select a workflow as shown in **Fig 1.3**. Select the desired solver and units of measurements as shown in **Fig. 1.4**.

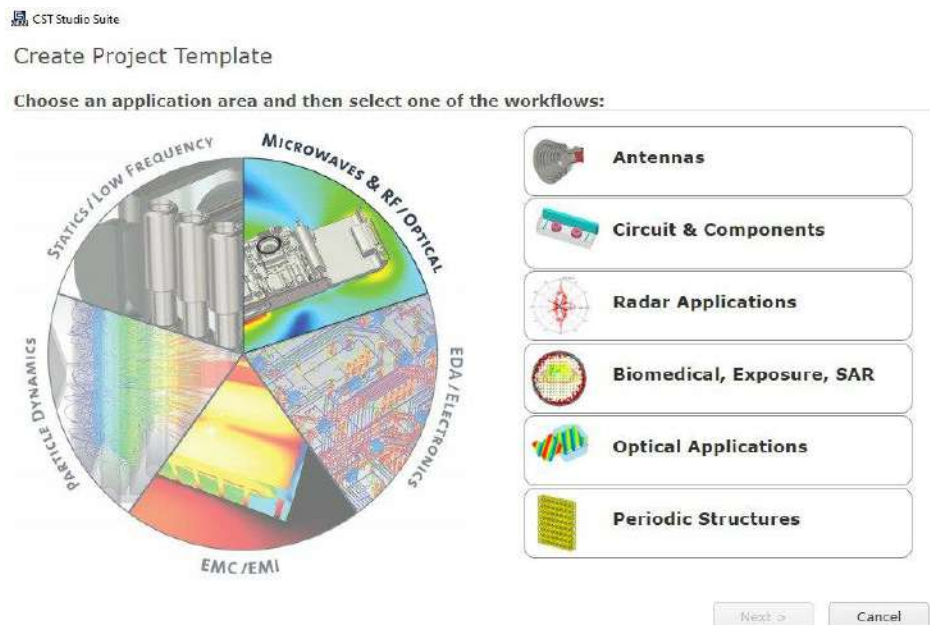


Fig. 1.2: Choosing a project template



Fig. 1.3: Selecting a workflow



Fig. 1.4: Selecting the desired solver

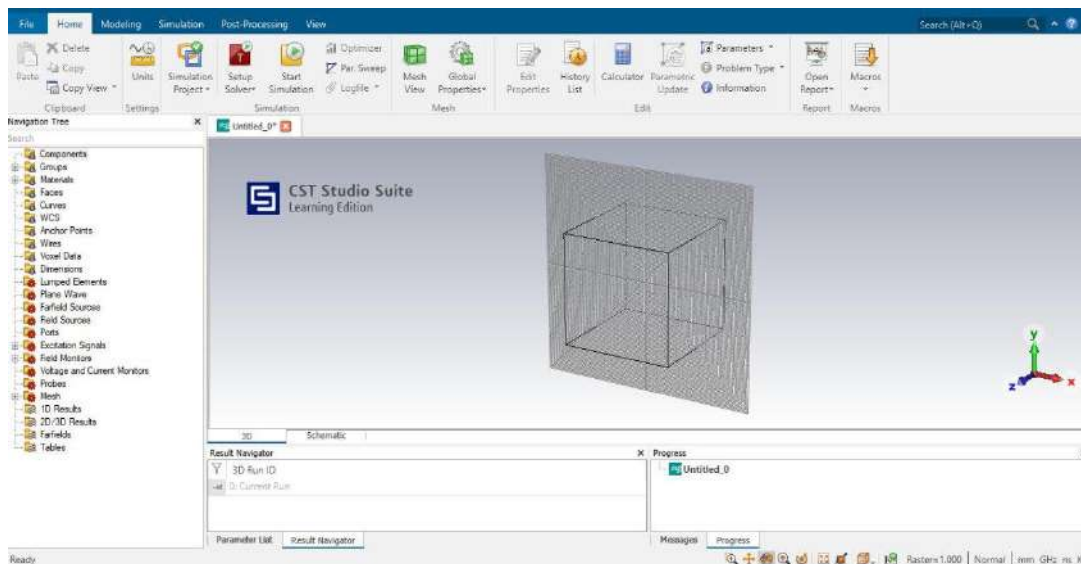


Fig. 1.5: Created canvas for project

The CST Toolbar shown in **Fig. 1.6** has all necessary tools for modelling and simulating antennas. The simulation results and plots can be obtained from the Navigation Tree shown in **Fig. 1.5**. Detailed procedure for each simulation is provided in the following experiments.

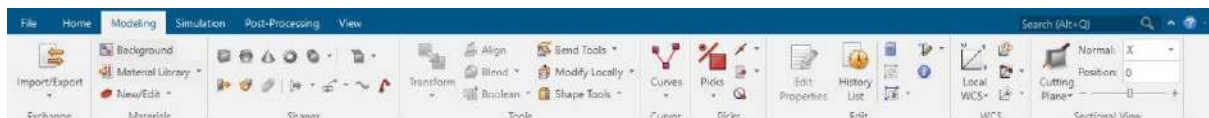


Fig. 1.6: CST Toolbar

Result

The familiarisation of CST Studio was completed.

EXPERIMENT NO 2

SIMULATION OF DIPOLE ANTENNA

Aim

To design and simulate a half wave dipole antenna at resonant frequency of 2.4 GHz and analyse its far-field parameters.

Tools used

CST Studio Suite 2022(Learning Edition)

Theory

Dipole antenna is considered to be the simplest practical antenna. It is a horizontal metallic rod which can be excited externally. Half-wavelength dipole antenna is one of the types of dipole antenna, in which the length of the antenna is half its wavelength at the frequency of operation. It can operate in the bandwidth of 3kHz-300GHz. It consists of two identical conductive metal rods which is fed externally at the centre of the space between two rods.

Working

A half-wave dipole antenna consists of two quarter-wavelength conductors placed end to end for a total length of approximately $L = \lambda/2$. The current distribution is that of a standing wave, approximately sinusoidal along the length of the dipole, with a node at each end and an antinode (peak current) at the centre (feed point):

$$I(z) = I_0 \cos \cos \omega t \cos \cos kz$$

where $k = 2\pi/\lambda$ and z runs from $-L/2$ to $L/2$. In the far field, this produces a radiation pattern whose electric field is given by

$$E_\theta = \frac{\zeta_0 I_0 \cos \cos \left(\frac{\pi}{2} \cos \cos \theta \right)}{2\pi r \sin \sin \theta} \sin \sin (\omega t - kr)$$

The directional factor $\cos[(\pi/2)\cos \theta]/\sin \theta$ is barely different from $\sin \theta$ applying to the short dipole, resulting in a very similar radiation pattern as noted above.

A numerical integration of the radiated power $|E^\theta|^2 / 2\zeta_0$ over all solid angle, as we did for the short dipole, obtains a value for the total power P_{total} radiated by the dipole with a current having a peak value of I_0 as in the form specified above. Dividing P_{total} by $4\pi R^2$ supplies the flux at a distance R averaged over all directions. Dividing the flux in the $\theta = 0$ direction (where it is at its peak) at distance R by that average flux, we find the directive gain to be 1.64. This can also be directly computed using the cosine integral:

$$G = \frac{4}{\text{Cin}(2\pi)} \approx 1.64(2 \cdot 15 \text{ dB})$$

Using P_{total}

$$P_{\text{total}} = \frac{1}{2} I_0^2 R_{\text{radiation}}$$

Radiation pattern can be obtained as:

$$R_{\text{radiation}} \approx 73.1 \Omega$$

Design

Resonant Frequency, $f=2.4\text{GHz}$ (given)

We know, $c = f \lambda$

Where c is the speed of light ($c=3 \times 10^8 \text{ m/s}$)

$$\text{Wavelength, } \lambda = \frac{c}{f}$$

Therefore, $\lambda = 0.125 \text{ m}$

$$\text{Length of the dipole, } L = \frac{143}{f(\text{MHz})}$$

Hence, $L = 59.5 \text{ mm}$

Radius of the wire, $R = 0.001 \lambda = 0.125 \text{ mm}$

Feeding gap, $g = L/200 = 0.2975 \text{ mm}$

Impedance $= 73 \Omega$

Procedure

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Wire/Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Draw the required antenna structure using 'Modelling tool'.
- iii. Enter the parameters in the parameter list as shown below:
 - a. $L=57.46 \text{ mm}$
 - b. Radius of the dipole, $r=0.3 \text{ mm}$
 - c. Feed gap, $g=2.88 \text{ mm}$
- iv. Select the face centre on either side of the gap → Exciting the antenna using discrete port after assigning the impedance as 73Ω .
- v. Set the field monitors → In that setup, set the fairfield monitor at 2.4 GHz .
- vi. Start the simulation.
- vii. Analyse the reflection coefficient(S_{11}) and the step impedance.
- viii. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- ix. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

Observations

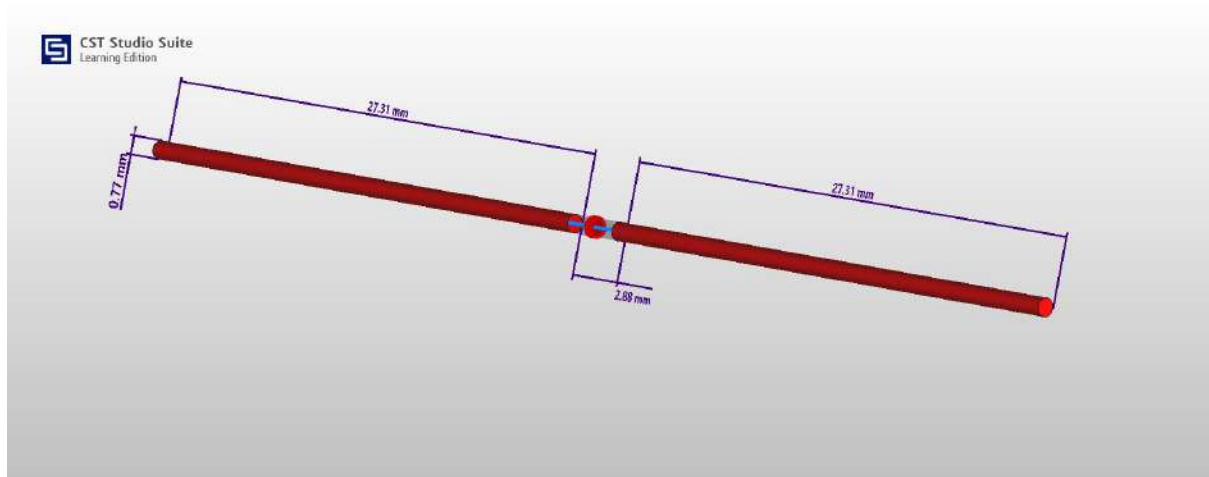


Fig. 2.1: Schematic of dipole antenna

s-parameters

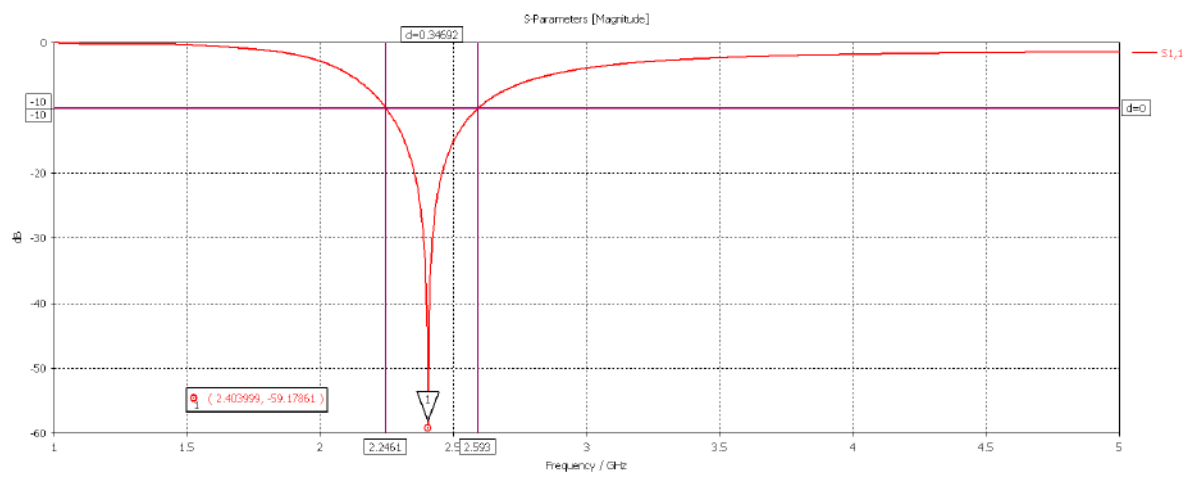


Fig. 2.2: s-parameter of dipole antenna

Return loss at 2.4GHz = 0.34692

VSWR

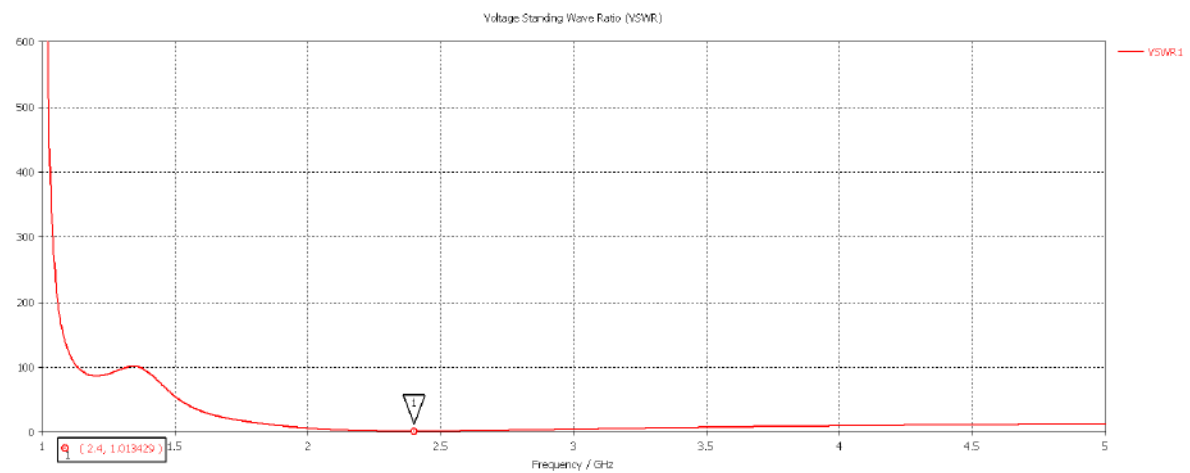


Fig. 2.3: VSWR of dipole antenna

VSWR at 2.4GHz = 1.013429

Radiation pattern

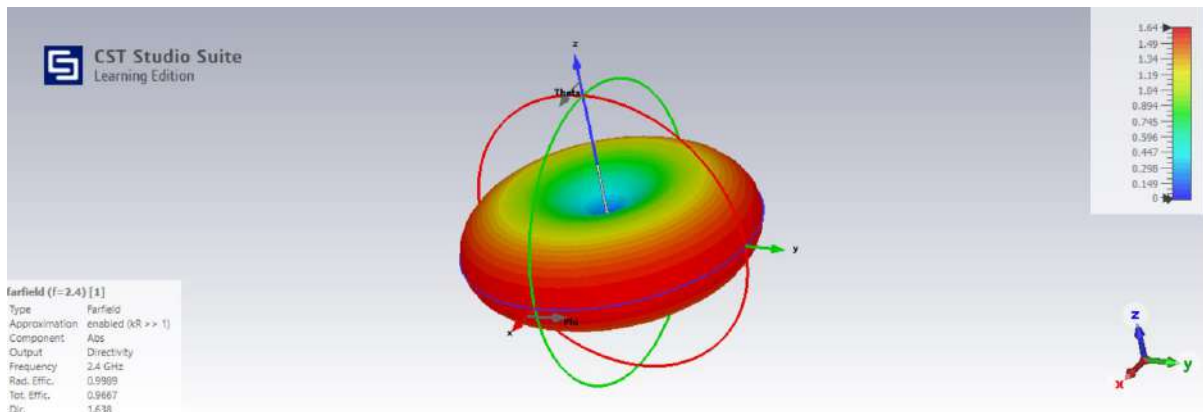
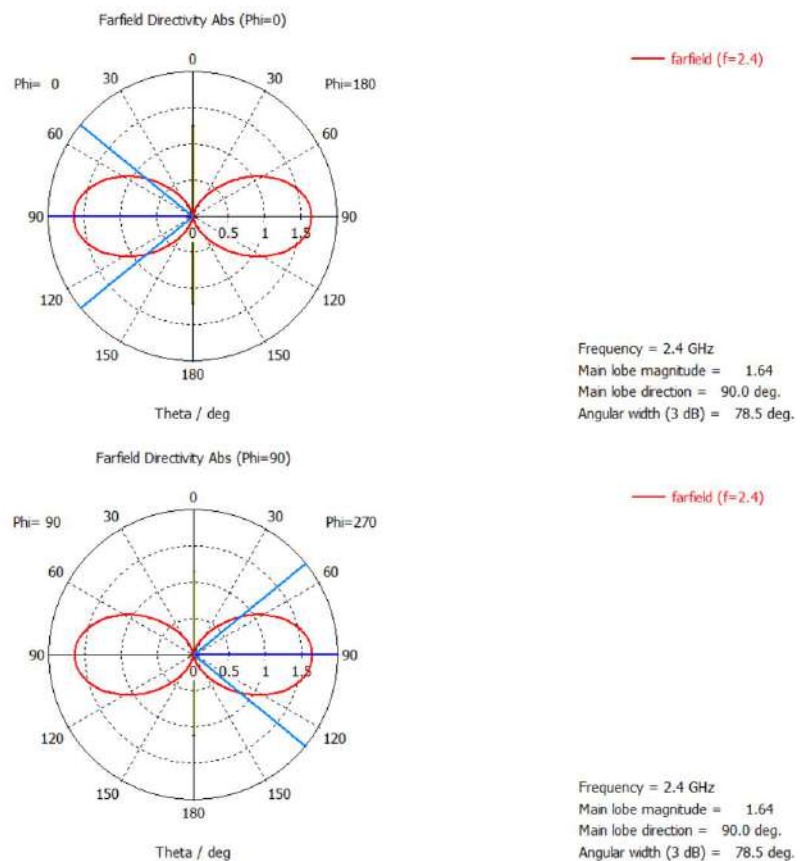


Fig. 2.4: 3D Radiation pattern of dipole antenna (Donut shaped)



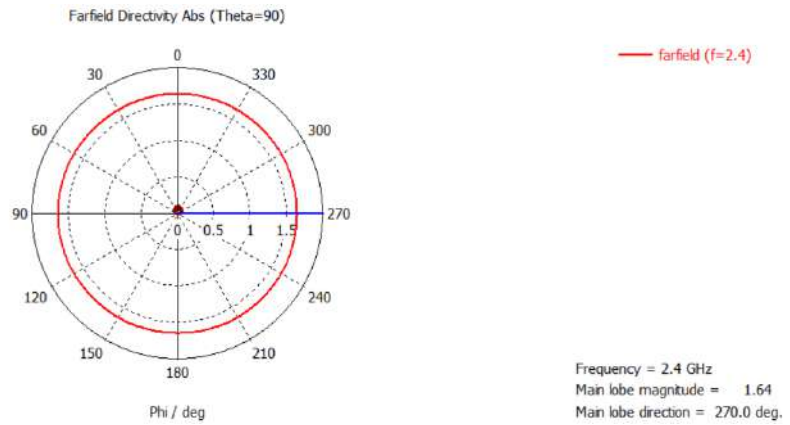


Fig.2.3: Polar patterns of dipole antenna

Result

Half-wavelength dipole was simulated and its far-field parameters were analysed. Obtained directivity, $D=1.633$ (2.132 dB).

EXPERIMENT NO 3

DESIGN AND SIMULATION OF MICROSTRIP PATCH ANTENNA USING MICROSTRIP FEED

Aim

To simulate a patch antenna with microstrip feed and analyse its far-field parameters.

Tools used

CST Studio Suite 2022(Learning Edition)

Theory

A patch antenna is a type of antenna with a low profile, which can be mounted on a surface. It consists of a planar rectangular, circular, triangular, or any geometrical sheet or "patch" of metal, mounted over a larger sheet of metal called a ground plane. They are the original type of microstrip antenna described by Howell in 1972; the two metal sheets together form a resonant piece of microstrip transmission line with a length of approximately one-half wavelength of the radio waves. The radiation mechanism arises from fringing fields along the radiating edges. The radiation at the edges causes the antenna to act slightly larger electrically than its physical dimensions, so in order for the antenna to be resonant, a length of microstrip transmission line slightly shorter than one-half the wavelength at the frequency is used. The patch antenna is mainly practical at microwave frequencies, at which wavelengths are short enough that the patches are conveniently small. It is widely used in portable wireless devices because of the ease of fabricating it on printed circuit boards. Multiple patch antennas on the same substrate called microstrip antennas, can be used to make high gain array antennas, and phased arrays in which the beam can be electronically steered. Radiate primarily because of the fringing fields between the patch

edge and the ground plane. The patch's radiation at the fringing fields results in a certain far-field radiation pattern. Maximum directivity in the direction perpendicular to the patch (broadside). Impedance is minimum (theoretically zero) in the middle of the patch and maximum at the edges. There is a point where the impedance is 50Ω somewhere along the resonant length.

A variant of the patch antenna commonly used in mobile phones is the shorted patch antenna, or planar inverted-F antenna (PIFA). In this antenna, one corner of the patch (or sometimes one edge) is grounded with a ground pin. This variant has better matching than the standard patch. Another variant of patch antenna with the partially etched ground plane, also known as printed monopole antenna, is a very versatile antenna for dual-band operations.

Design

Dielectric constant, $\epsilon_r = 4.3$

Dielectric used = FR4(loss Free)

Operating frequency, $f_r = 2.4\text{GHz}$

Height of the substrate, $h = 1.6\text{mm}$

Width of the patch, $W = \frac{1}{2f_r\sqrt{\mu_0\epsilon_0}}\sqrt{\frac{2}{\epsilon_r+1}} = 0.0383\text{m} = 38.3\text{mm}$

The effective dielectric constant of the microstrip antenna,

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-\frac{1}{2}} = 2.7$$

$$\text{Extension of the length } L, \frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{reff} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{reff} - 0.258) \left(\frac{W}{h} + 0.8 \right)} = 0.495$$

$$\text{The actual length of the patch, } L = \frac{1}{2f_r \sqrt{\epsilon_{reff} \mu_0 \epsilon_0}} - 2 \Delta L = 29.77$$

Length (L_g) and Width (W_g) of the ground plane,

$$L_g = 6h + L = 39.37$$

$$W_g = 6h + W = 47.99$$

Width of the feed line = 3.1mm

Let, height of the ground plane, $t = 0.035\text{mm}$.

Procedure

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Draw the required antenna structure using 'Modelling tool'.
- iii. Define parameters in the parameter list.
 - a. $L = 29.77\text{mm}$
 - b. $W = 38.39\text{mm}$
 - c. $L_g = 39.37\text{mm}$
 - d. $W_g = 47.99\text{mm}$
 - e. $h = 1.6\text{mm}$
 - f. $W_s = 47.99\text{mm}$
 - g. $L_s = 39.37\text{mm}$
 - h. $t = 0.035\text{mm}$
 - i. $W_f = 3.1\text{mm}$
- iv. Connect the ports → Connect the waveguide ports by calculating the size of the port using the 'Calculate port extension coefficient' in the 'Solvers' option inside the 'Macros' tab.
- v. Set the field monitors → In that setup, set the fairfield monitor at 2.4GHz.
- vi. Start the simulation.
- vii. Analyse the reflection coefficient(S_{11}) and the step impedance.
- viii. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- ix. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

Observations

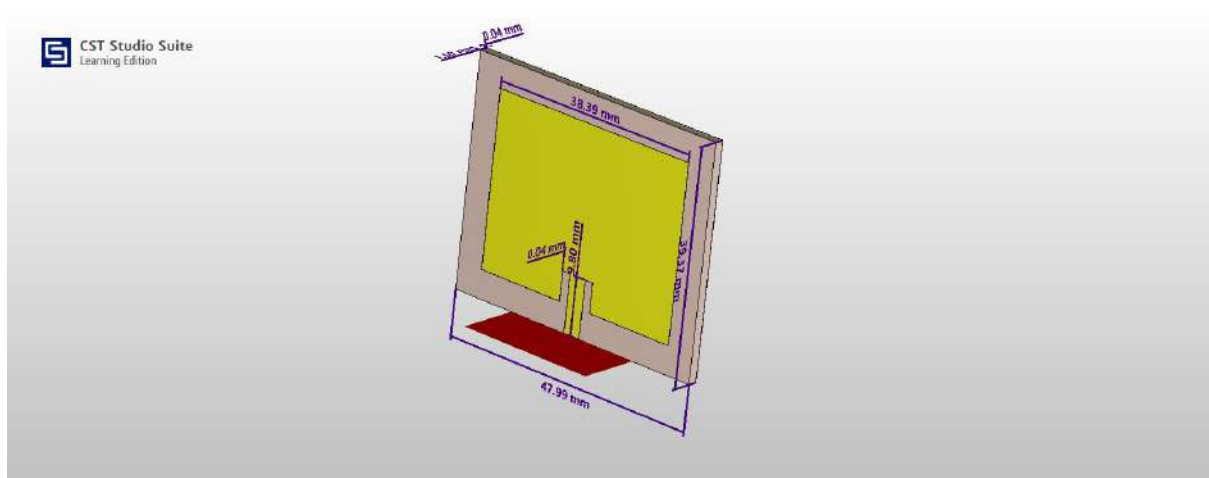


Fig. 3.1: Schematic of patch antenna

s-parameters

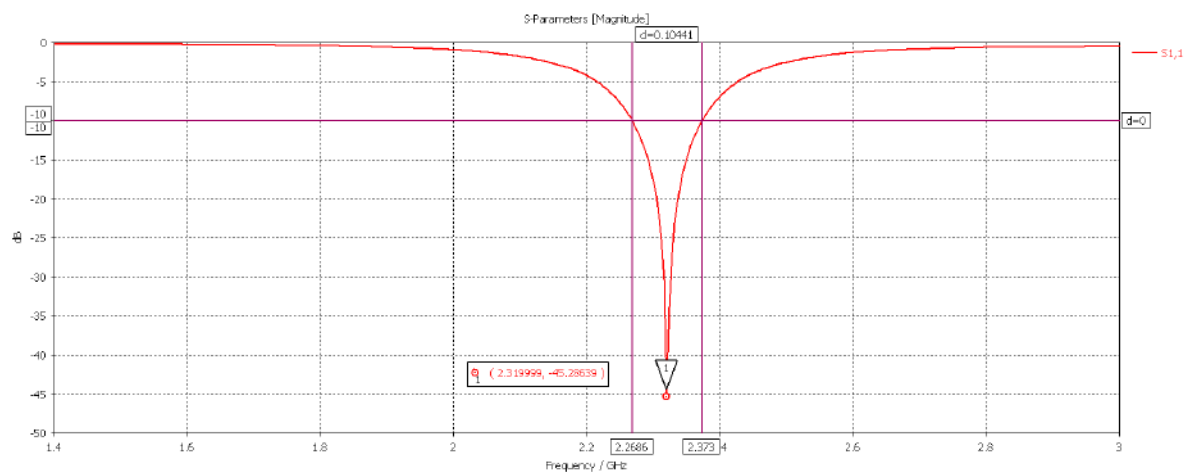


Fig. 3.4: s-parameter of patch antenna

VSWR

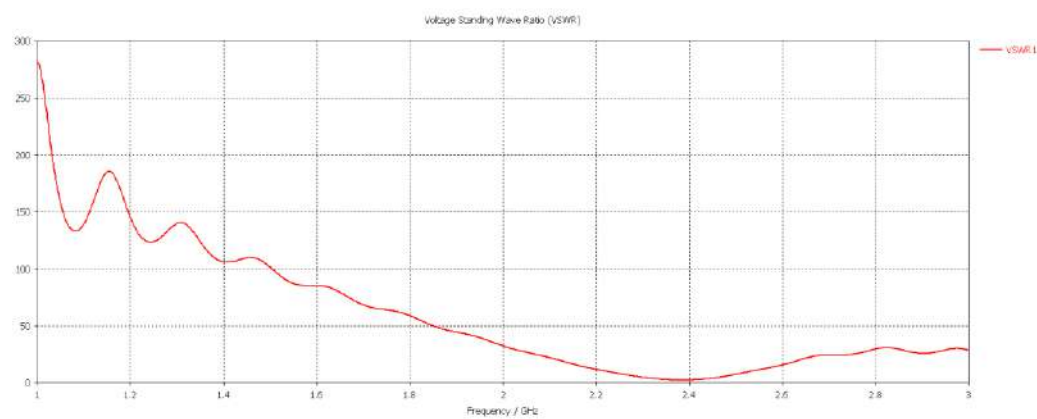


Fig. 3.4: VSWR of patch antenna

VSWR at 2.4GHz = 3.401

Radiation pattern

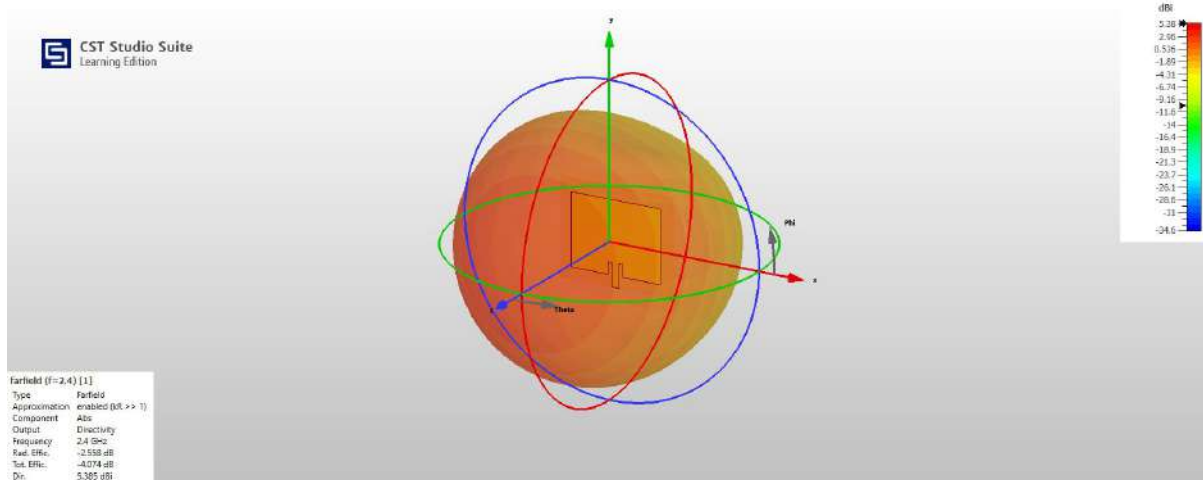
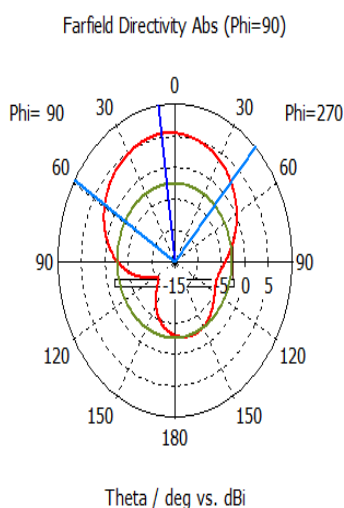
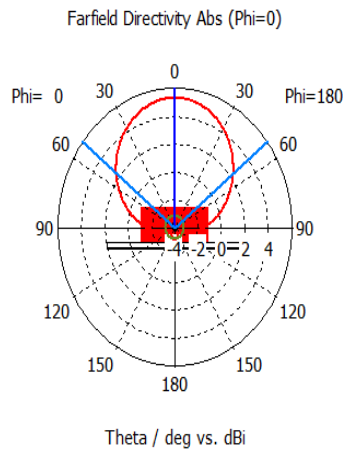


Fig. 3.3: 3D radiation pattern of patch antenna



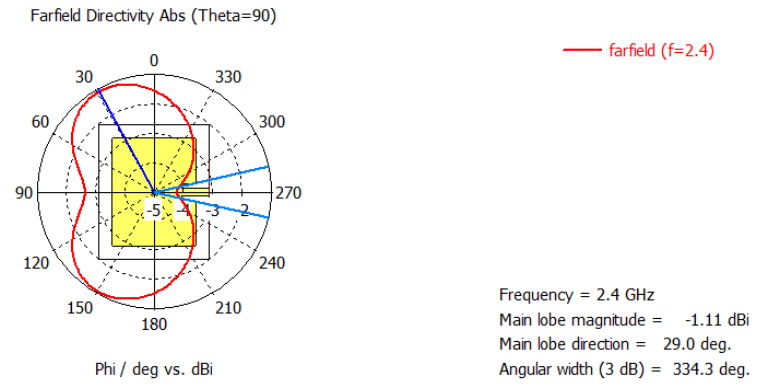


Fig. 3.3: 1D radiation pattern of patch antenna

Result

Patch antenna with microstrip feed was simulated and its far-field parameters were analysed. Obtained directivity, $D = 3.455$ (5.385 dB).

EXPERIMENT NO 4

SIMULATION OF MICROSTRIP ANTENNA ARRAY

Aim

To simulate an antenna array with corporate microstrip feed and analyse its far-field parameters.

Tools used

CST Studio Suite 2022(Learning Edition)

Theory

An antenna array (or array antenna) is a set of multiple connected antennas which work together as a single antenna, to transmit or receive radio waves. The individual antennas (called elements) are usually connected to a single receiver or transmitter by feedlines that feed the power to the elements in a specific phase relationship. The radio waves radiated by each individual antenna combine and superpose, adding together (interfering constructively) to enhance the power radiated in desired directions, and cancelling (interfering destructively) to reduce the power radiated in other directions. Similarly, when used for receiving, the separate radio frequency currents from the individual antennas combine in the receiver with the correct phase relationship to enhance signals received from the desired directions and cancel signals from undesired directions. More sophisticated array antennas may have multiple transmitter or receiver modules, each connected to a separate antenna element or group of elements. An antenna array can achieve higher gain (directivity), that is a narrower beam of radio waves, than could be achieved by a single element. In general, the larger the number of individual antenna elements used, the higher the gain and the narrower the beam. Some antenna arrays (such as military phased array radars) are composed of thousands of individual antennas. Arrays can be used to achieve higher gain, to give path diversity (also called MIMO) which increases communication reliability, to cancel interference from specific directions, to steer the radio beam electronically to point in different directions, and for radio direction finding (RDF). The term antenna array most commonly means a driven array consisting of multiple identical driven elements all connected to the receiver or transmitter. A parasitic array consists of a single driven element connected to the feedline, and other elements which are not, called parasitic elements. It is usually another name for a Yagi–Uda antenna. A phased array usually means an electronically scanned array; a driven array antenna in which each individual element is connected to the transmitter or receiver through a phase shifter controlled by a computer. The beam of radio waves can be steered electronically to point instantly in any direction over a wide angle, without moving the antennas. However, the term "phased array" is sometimes used to mean an ordinary array antenna.

Design

Dielectric used = FR-4(loss free)

Dielectric constant, $\epsilon_r = 4.3$

Operating frequency, $f_r = 5.6\text{GHz}$

Height of the substrate = 1.6mm

$$\text{Width of the Patch, } W = \frac{1}{2fr\sqrt{\mu_0\epsilon_0}} \sqrt{\frac{2}{\epsilon_r + 1}} = 16.33\text{mm}$$

The effective dielectric constant of the microstrip antenna,

$$\epsilon_{\text{reff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-0.5} = 2.43$$

Extension of the length ΔL ,

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{\text{reff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{reff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} = 0.7885\text{mm}$$

$$\text{The actual length of the patch, } L = \frac{1}{2fr\sqrt{\epsilon_{\text{reff}}\mu_0\epsilon_0}} - 2\Delta L = 12.33\text{mm}$$

Length (L_g) and Width (W_g) of the ground plane,

$$L_g = 6h + L = 21.93\text{mm}$$

$$W_g = 6h + L = 26.04\text{mm}$$

Width of the feed line = 3.137mm

Let height of the ground plane, $t = 0.035\text{mm}$

Gap between each microstrip patch antenna = $\lambda/2$

Procedure

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Draw the required antenna structure using 'Modelling tool'.
- iii. Connect the ports → Connect the waveguide ports by calculating the size of the port using the 'Calculate port extension coefficient' in the 'Solvers' option inside the 'Macros' tab.
- iv. Set the field monitors → In that setup, set the fairfield monitor at 5.6GHz.
- v. Start the simulation.
- vi. Analyse the reflection coefficient(S11) and the step impedance.
- vii. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- viii. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

Observations

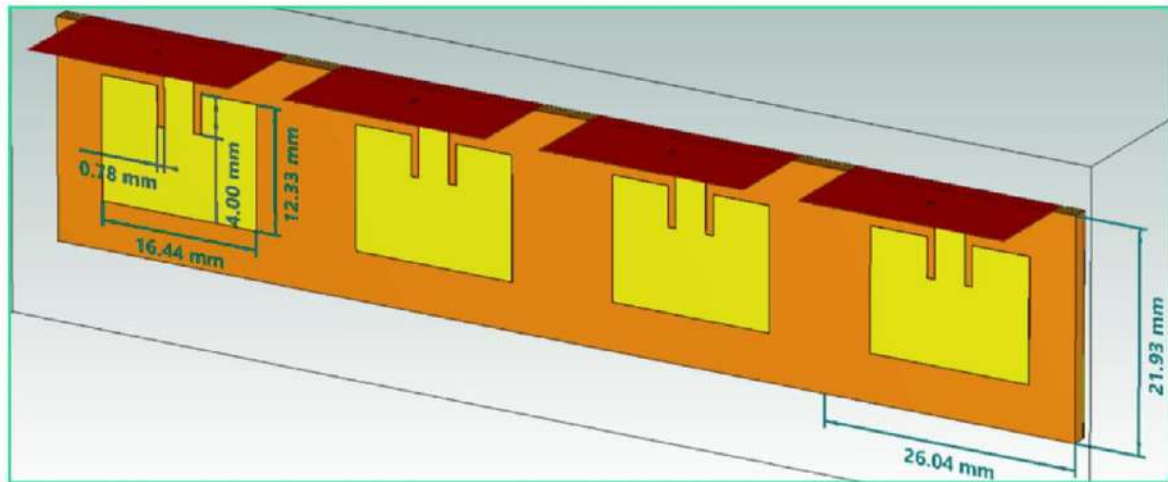


Fig. 4.1: Schematic of 1×4 antenna array

Results

s-parameters

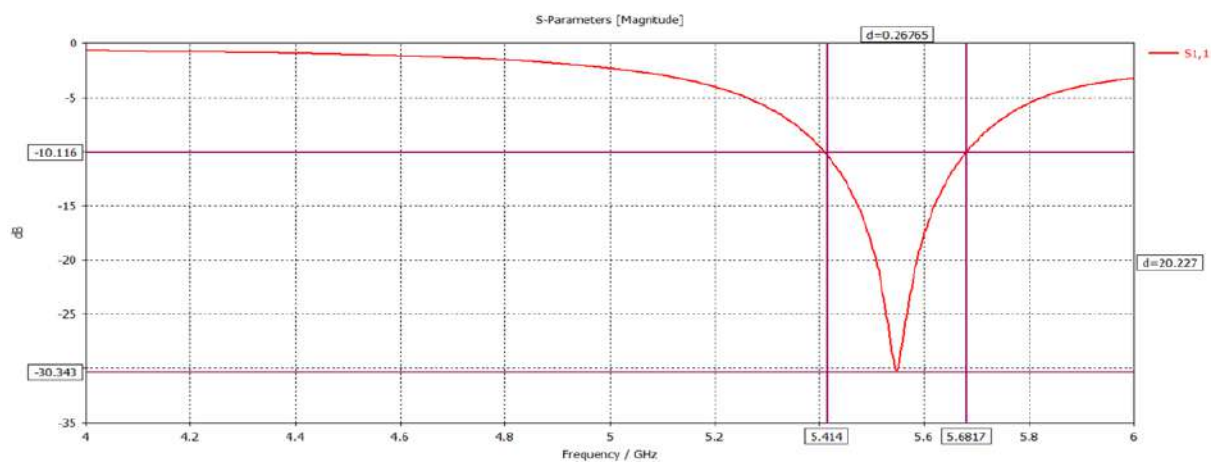


Fig. 4.2: s-parameter of 1×4 antenna array

Radiation pattern

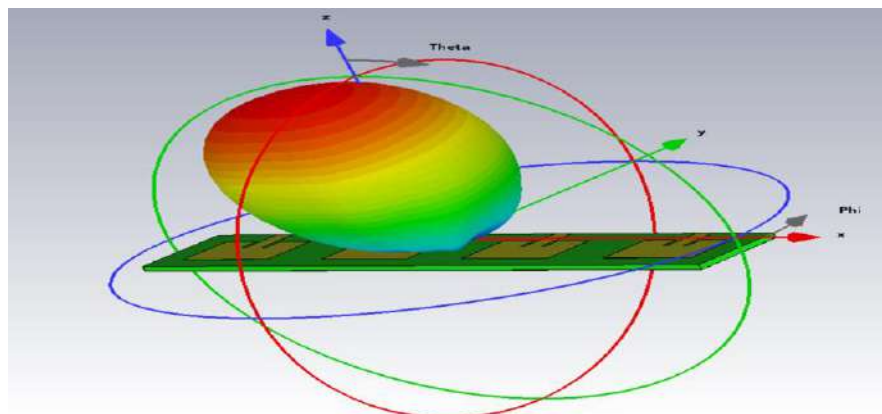


Fig. 4.4: 3D radiation pattern of 1×4 antenna array

Far field cuts

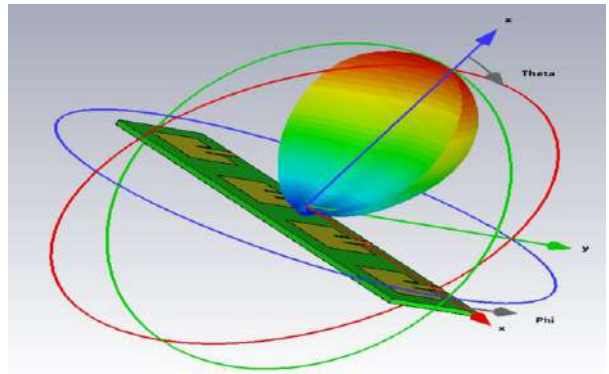
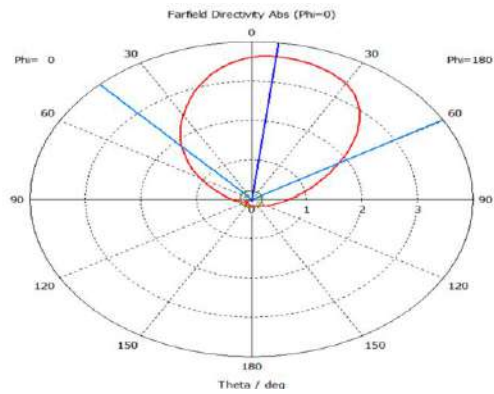


Figure 4.4.1 XZ Plane [OR] $\Phi = 0^\circ$ Plane

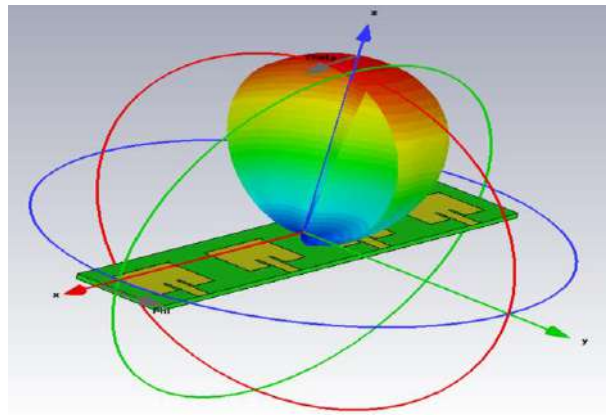
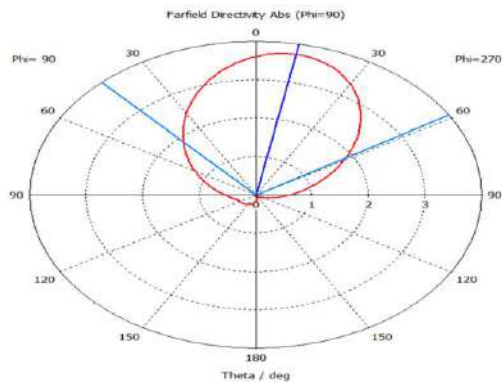


Figure 4.4.2 YZ Plane [OR] $\Phi = 90^\circ$ Plane

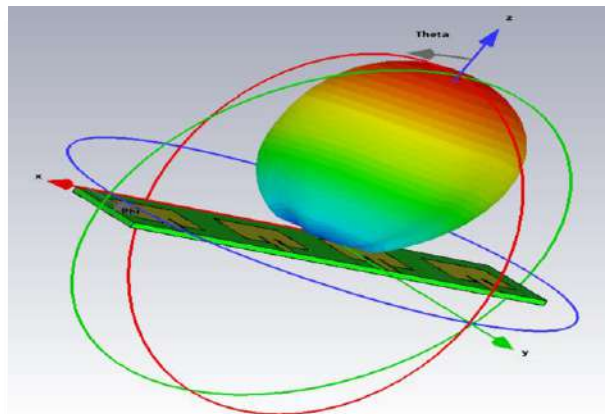
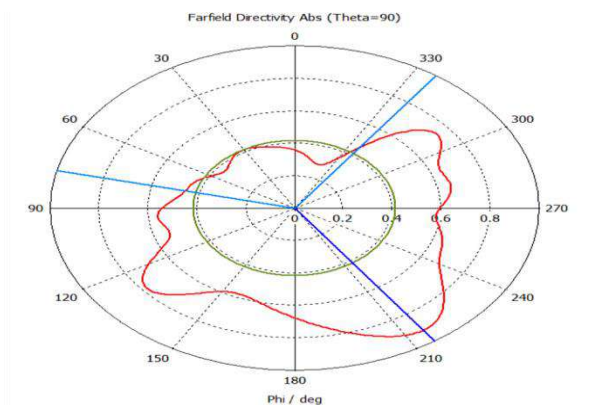


Figure 4.4.3 XY Plane [OR] $\theta = 90^\circ$ Plane

1×4 Antenna array was simulated and its far-field parameters were analysed.

EXPERIMENT NO 5

FOLDED DIPOLE ANTENNA

AIM

To design and simulate a folded dipole Antenna for 2.4GHz

DESIGN

Material used = Copper (Annealed)

Material Type = Lossy Metal

$$\begin{aligned}\lambda &= c/f \\ &= (3 \times 10^8)/(2400)\text{MHz} \\ &= 0.125\text{m} = 125\text{mm}\end{aligned}$$

$$\begin{aligned}\text{Length (L)} &= 143/f = 143/2400(\text{MHz}) \\ &= 59.58\text{mm}\end{aligned}$$

$$\begin{aligned}\text{Radius(R)} &= 0.001 \times \text{Lambda} \\ &= 0.001 \times 125 \\ &= 0.125\text{mm}\end{aligned}$$

$$\begin{aligned}\text{Gap(g)} &= L/200 \\ &= 59.58/200 \\ &= 0.29\text{mm}\end{aligned}$$

PROCEDURE

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Draw the required antenna structure using 'Modelling tool'.
- iii. Draw the structure solid tool as prescribed. Use the transform tool to replicate the same structure. Adjust parameters and maintain distance between them.
- iv. Design a small cylinder and place it in perpendicular to the main line at the middle of both lines and place it at some distance from one end.

- v. Use the 'Loft' tool to join the ends of the main line and the small cylindrical tool. Replicate the same on the other side by using the transform tool, select mirror, copy and adjust axis parameters for fitting correctly.
- vi. Add all solids by using the 'Boolean Add' tool and make it one solid.
- vii. Cut a small cylindrical shape out of one main line and make that gap vacuum. By using the pick tool and selecting faces and creating ports.
- viii. Start simulation.
- ix. Analyse the reflection coefficient(S11) and the step impedance.
- x. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- xi. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

OBSERVATIONS

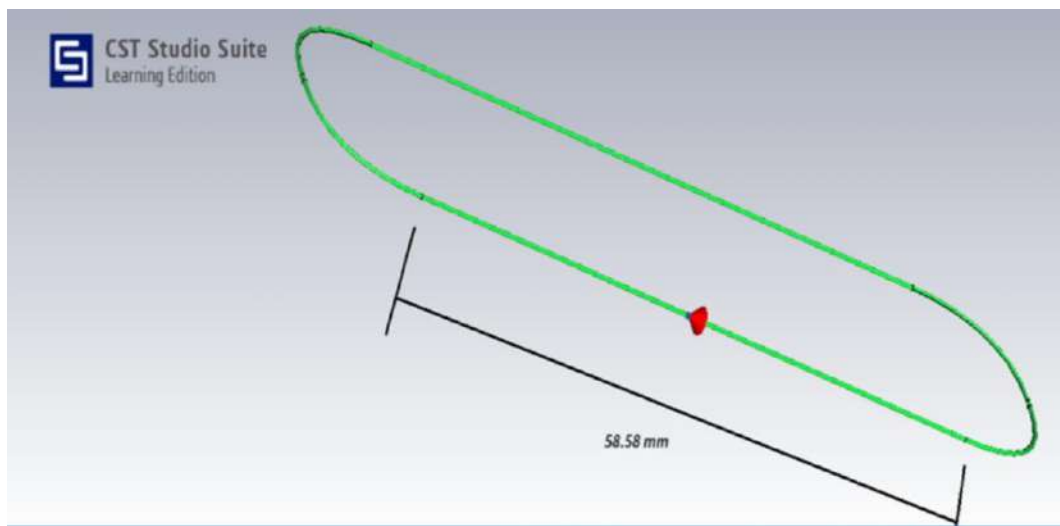


Figure 5.1 Schematic of the Folded Dipole Antenna

RESULTS

Reflection Characteristics(S11)

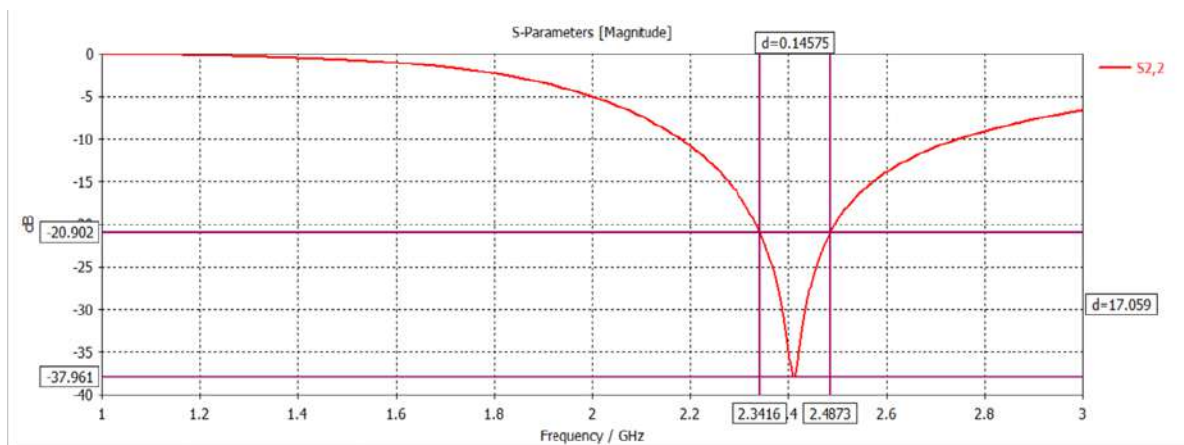


Figure 5.2 Reflection Characteristics of Folded Dipole Antenna with resonant frequency 2.4GHz

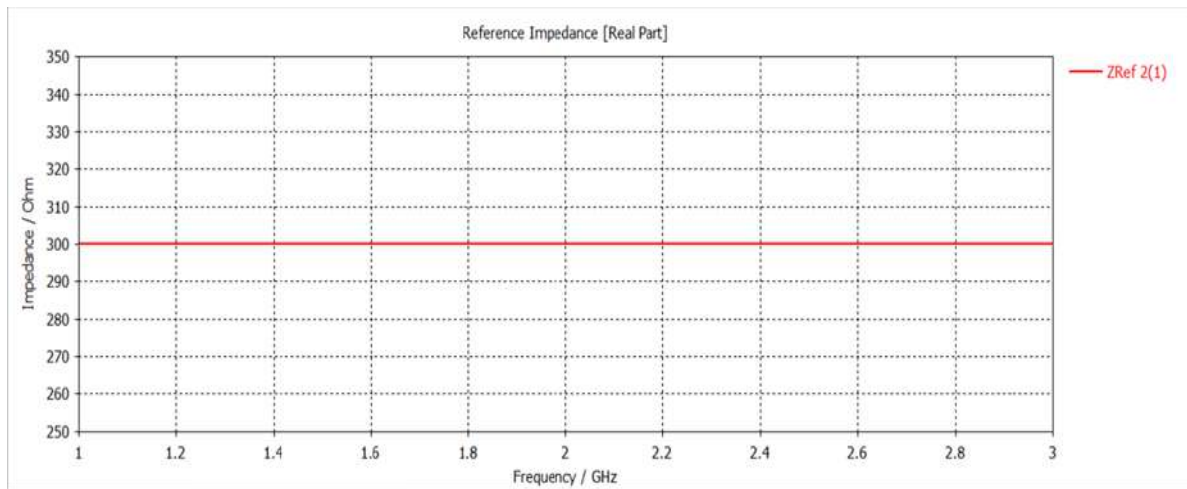


Figure 5.3 Reference Impedance plot

Far field Plot

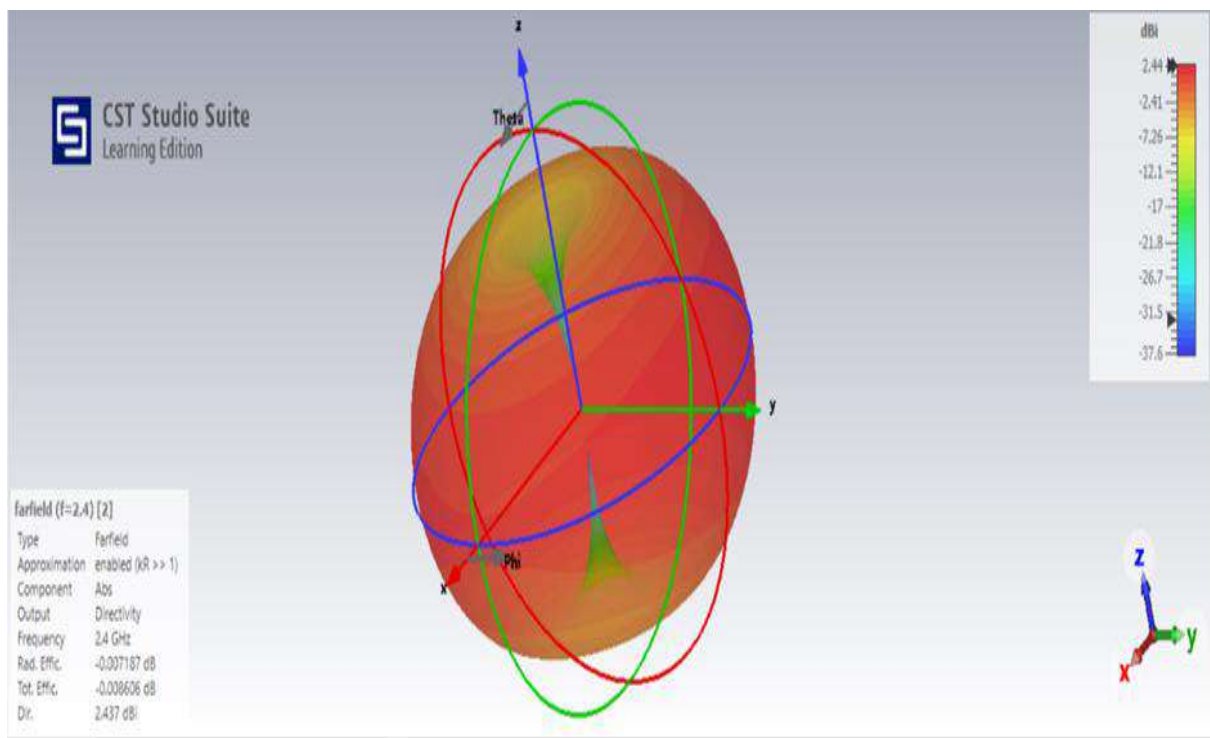
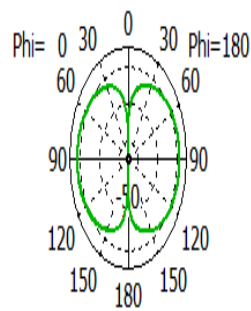


Figure 5.4 3-Dimensional Radiation Pattern

Farfield Directivity Abs (Phi=0)



Theta / deg vs. dBi

— farfield (f=2.4)

— farfield (f=2.4) (L=56.6667)

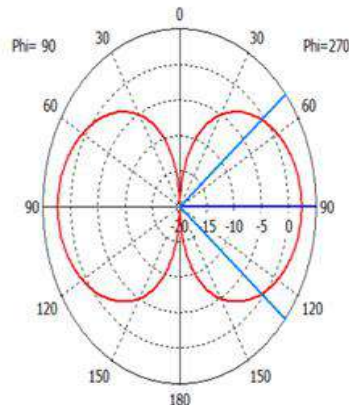
Frequency = 2.4 GHz

Main lobe magnitude = 2.24 dBi

Main lobe direction = 90.0 deg.

Angular width (3 dB) = 79.1 deg.

Farfield Directivity Abs (Phi=90)



Theta / deg vs. dBi

— farfield (f=2.4)

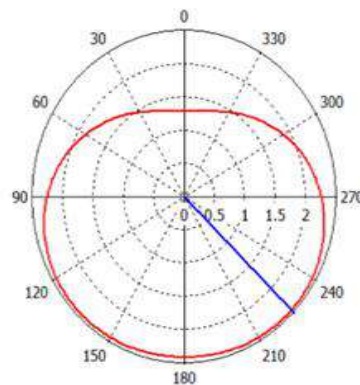
Frequency = 2.4 GHz

Main lobe magnitude = 2.25 dBi

Main lobe direction = 90.0 deg.

Angular width (3 dB) = 78.8 deg.

Farfield Directivity Abs (Theta=90)



Phi / deg vs. dBi

— farfield (f=2.4)

Frequency = 2.4 GHz

Main lobe magnitude = 2.44 dBi

Main lobe direction = 226.0 deg.

Figure 5.5 2 Dimensional cuts about different planes

Designed, simulated and studied the resonant frequency and the radiation pattern of the microstrip patch antenna.

EXPERIMENT NO 6

DESIGN AND SIMULATION OF PLANAR DIPOLE ANTENNA

AIM

To design and simulate a planar dipole antenna with resonant frequency 2.4GHz.

DESIGN

Dielectric = Rogers RT duroid 5880(Lossy)

Dielectric constant, $\epsilon_r = 2.2$

Operating frequency, $f_r = 5.6\text{GHz}$

Height of the substrate = 1.6mm

Width of the substrate = 10mm

Length of the substrate = 80mm

Length of the dipole = 0.001λ

PROCEDURE

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Draw the required antenna structure using 'Modelling tool'.
- iii. Draw the structure solid tool as prescribed. Use the transform tool to replicate the same structure. Adjust parameters and maintain distance between them.
- iv. Design a small cylinder and place it in perpendicular to the main line at the middle of both lines and place it at some distance from one end.
- v. Use the 'Loft' tool to join the ends of the main line and the small cylindrical tool. Replicate the same on the other side by using the transform tool, select mirror, copy and adjust axis parameters for fitting correctly.
- vi. Add all solids by using the 'Boolean Add' tool and make it one solid.
- vii. Cut a small cylindrical shape out of one main line and make that gap vacuum. By using the pick tool and selecting faces and creating ports.

- viii. Start Simulation.
- ix. Analyse the reflection coefficient(S11) and the step impedance.
- x. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- xi. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

OBSERVATIONS

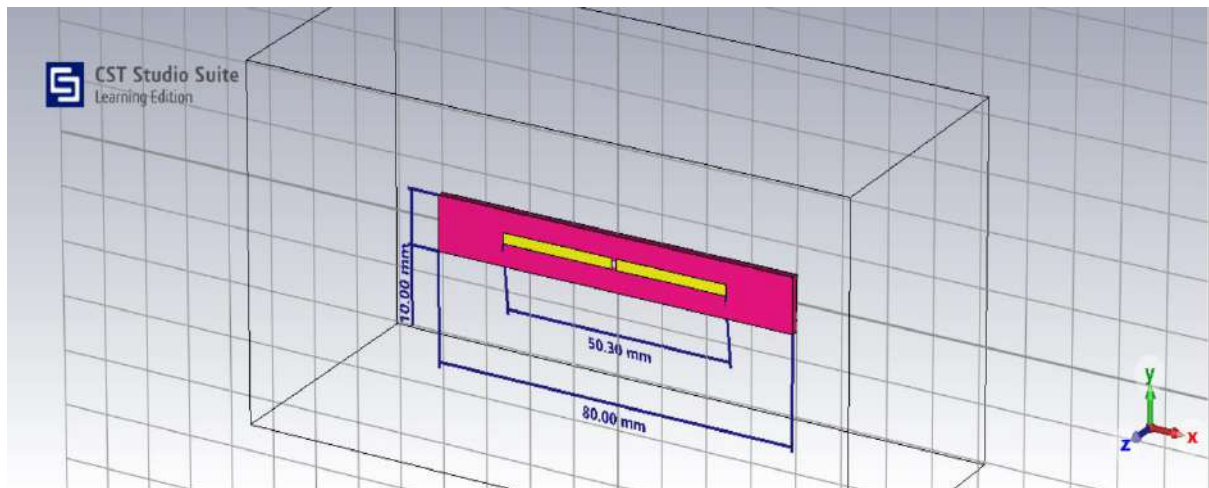


Figure 6.1 Schematic of Planar Dipole Antenna

RESULTS

Reflection Characteristics(S11)

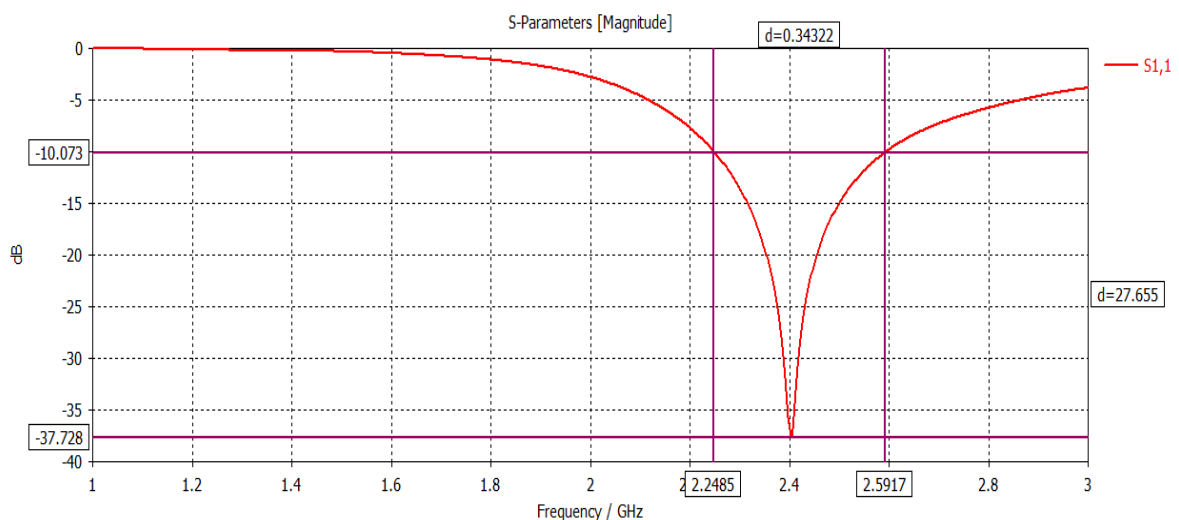


Figure 6.2 Reflection Characteristics of Planar antenna with resonant frequency 2.4 GHz

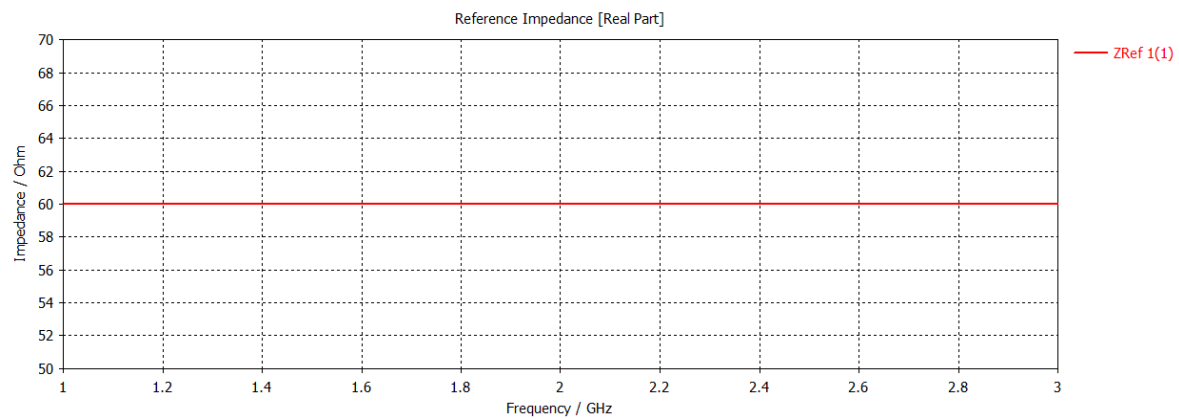


Figure 6.3 Reference Impedance Plot

Farfield Plot

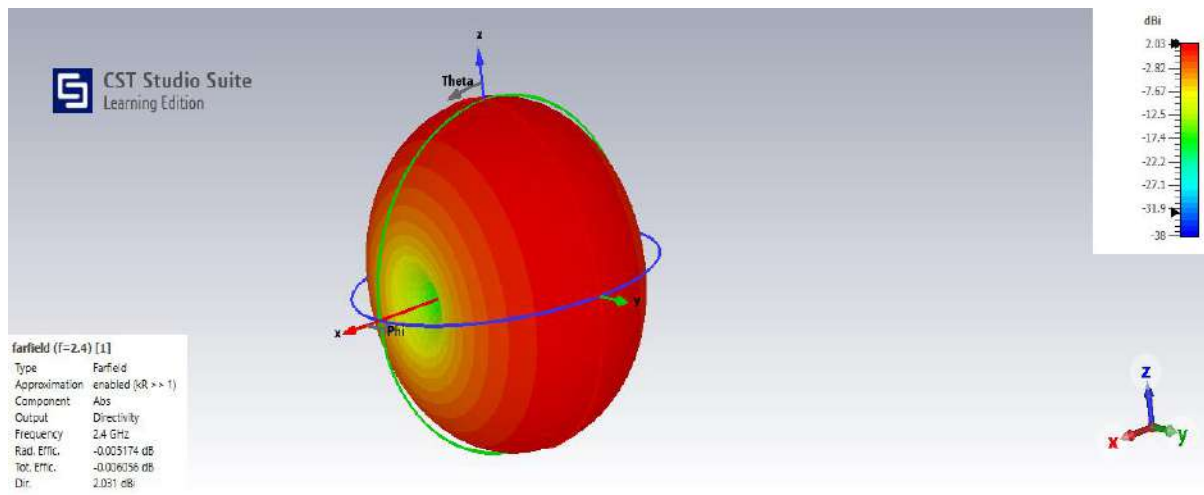
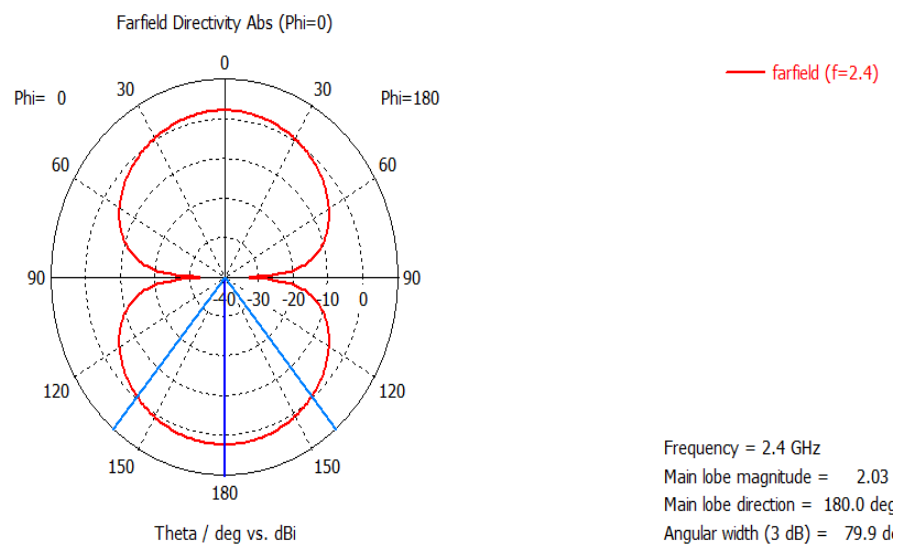
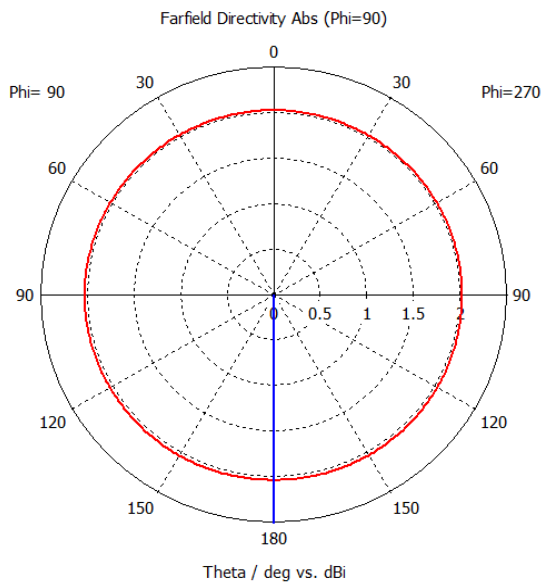


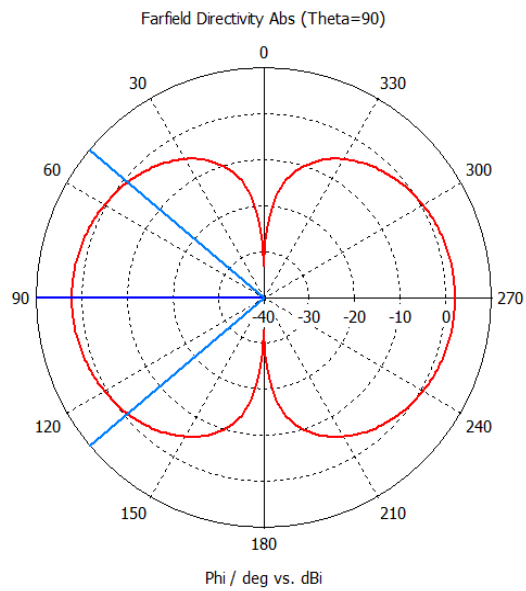
Figure 6.4 3 – Dimensional radiation pattern





— farfield (f=2.4)

Frequency = 2.4 GHz
Main lobe magnitude = 2.03 dBi
Main lobe direction = 180.0 deg.



— farfield (f=2.4)

Frequency = 2.4 GHz
Main lobe magnitude = 2.02 dBi
Main lobe direction = 90.0 deg.
Angular width (3 dB) = 80.2 deg.

Figure 6.5 2 Dimensional Cuts

EXPERIMENT NO 7

DESIGN AND SIMULATION OF PATCH ANTENNA USING COAXIAL FEED

AIM

To design and simulate a patch antenna with coaxial feed for a resonant frequency of 5.6GHz.

DESIGN

Dielectric used = FR-4(loss free)

Dielectric constant, $\epsilon_r = 4.3$

Operating frequency, $f_r = 5.6\text{GHz}$

Height of the substrate = 1.6mm

$$\text{Width of the Patch, } W = \frac{1}{2fr\sqrt{\mu_0\epsilon_0}} \sqrt{\frac{2}{\epsilon_r + 1}} = 16.33\text{mm}$$

The effective dielectric constant of the microstrip antenna,

$$\epsilon_{\text{reff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W}\right)^{-1} = 2.43$$

Extension of the length ΔL ,

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{\text{reff}} + 0.3) \left(\frac{W}{h} + 0.264\right)}{(\epsilon_{\text{reff}} - 0.258) \left(\frac{W}{h} + 0.8\right)} = 0.7885\text{mm}$$

$$\text{The actual length of the patch, } L = \frac{1}{2fr\sqrt{\epsilon_{\text{reff}}\mu_0\epsilon_0}} - 2\Delta L = 12.33\text{mm}$$

Length (L_g) and Width (W_g) of the ground plane,

$$L_g = 6h + L = 21.93\text{mm}$$

$$W_g = 6h + L = 26.04\text{mm}$$

Let height of the ground plane, $t = 0.035\text{mm}$

Diameter of outerpin, $D = 4.7\text{mm}$

Diameter of inner pin, $d = 1.4\text{mm}$

Coaxial feed coordinates, (Xf, Yf) :

$$X_f = \frac{L}{\sqrt[2]{\epsilon_{eff}}} \quad Y_f = \frac{w}{2}$$

PROCEDURE

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Calculate the radius of the coaxial feed using 'Calculate analytical Line impedance' in the 'Calculate' option inside the 'Macros' tab.
- iii. Draw the required antenna structure using 'Modelling tool'.
- iv. Assign the port using 'Pick Face' option, then click 'Apply' in the 'Waveguide Port' tab
- v. Set the field monitors → In that setup, set the farfield monitor at 5.6GHz.
- vi. Start the simulation.
- vii. Analyse the reflection coefficient(S11) and the step impedance. Change values of (Xf,Yf) for desired S11
- viii. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- ix. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

OBSERVATIONS

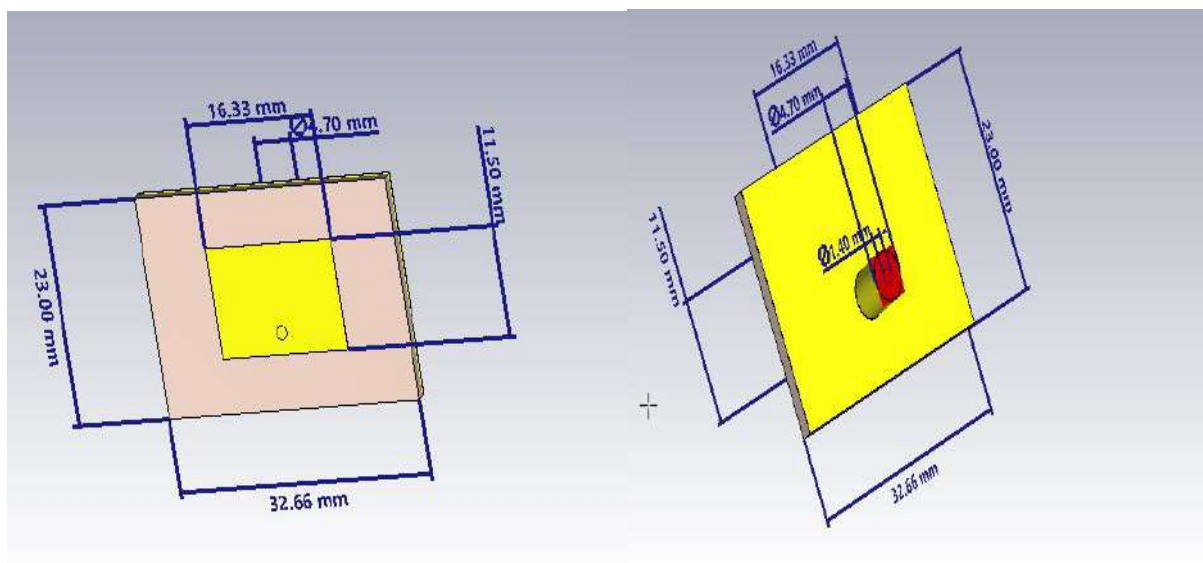


Figure 7.1 Schematic of Patch antenna using Coaxial Feed

RESULTS

Reflection Characteristics

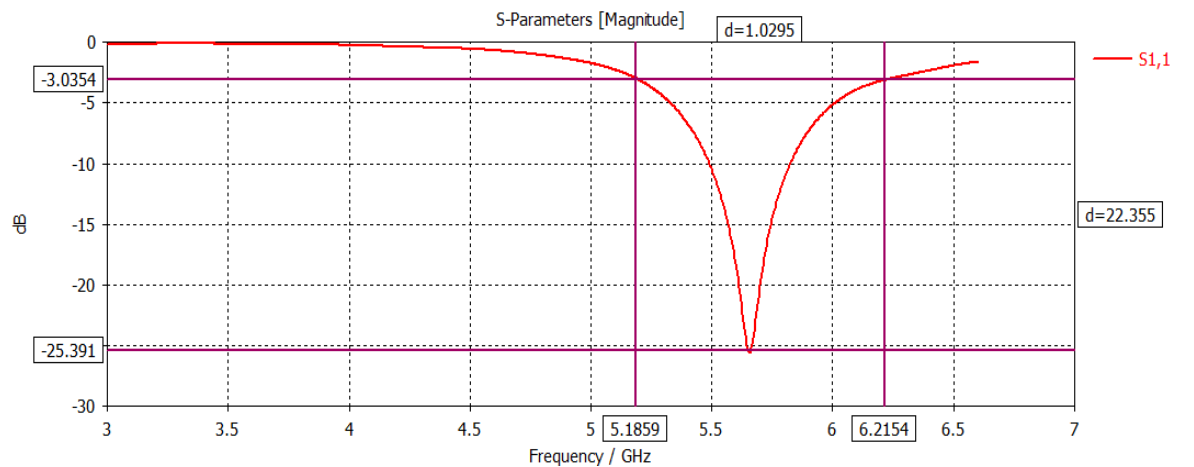


Figure 7.2 Reflection Characteristics of Patch Antenna with resonant frequency 5.6GHz

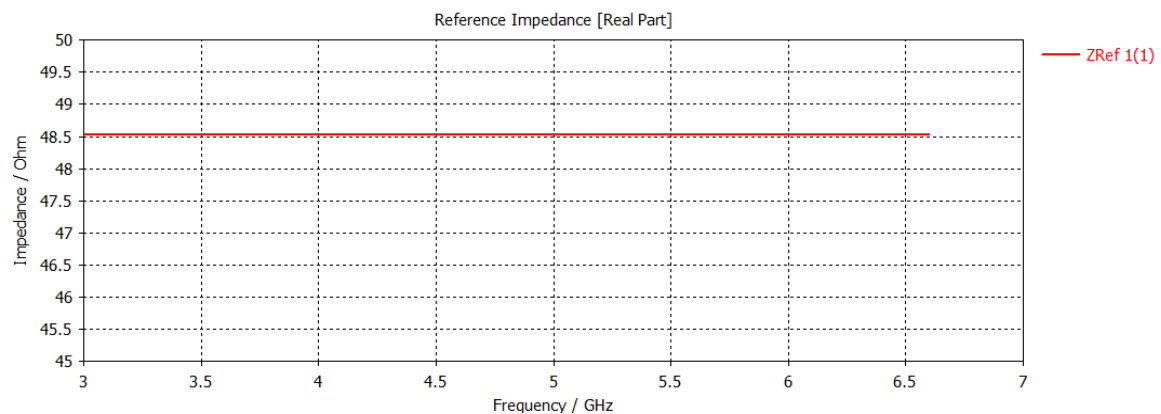


Figure 7.3 Reference Impedance plot

Far field Plot

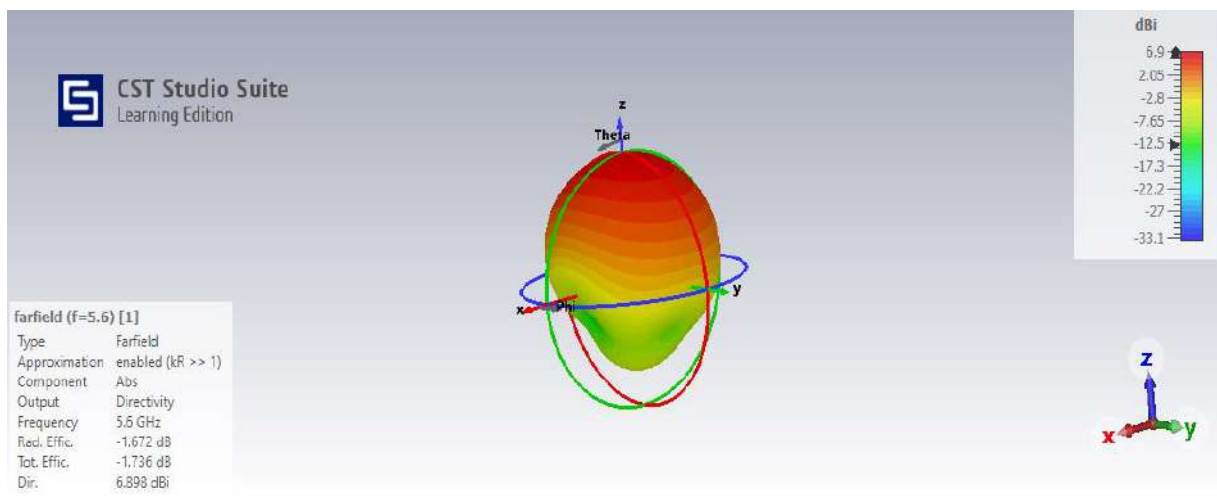


Figure 7.4 3 Dimensional Radiation Pattern

Far field cuts

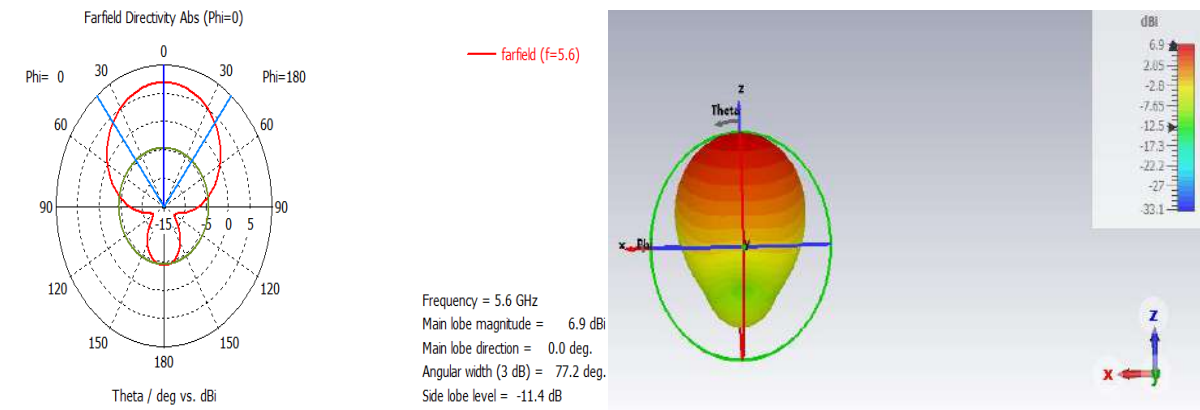


Figure 7.5.1 XZ Plane [OR] $\Phi = 0^\circ$ Plane

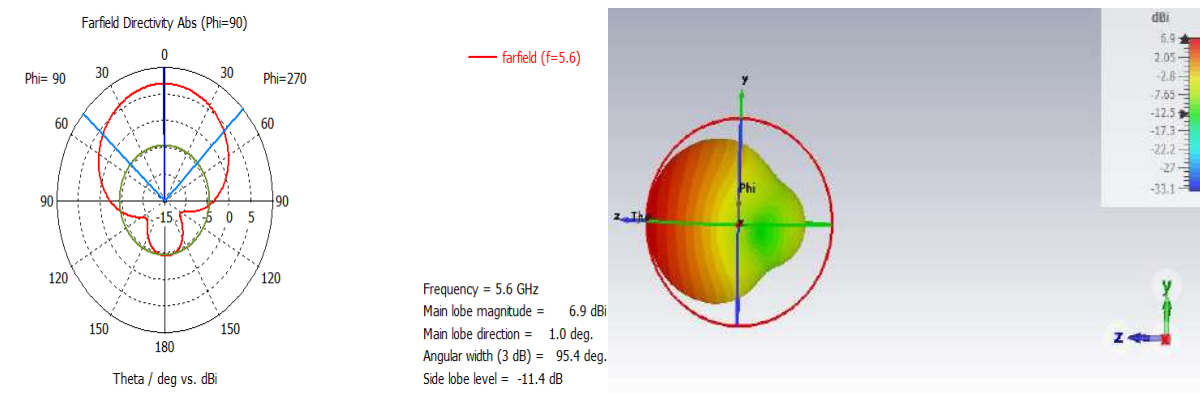


Figure 7.5.2 YZ Plane [OR] $\Phi = 90^\circ$ Plane

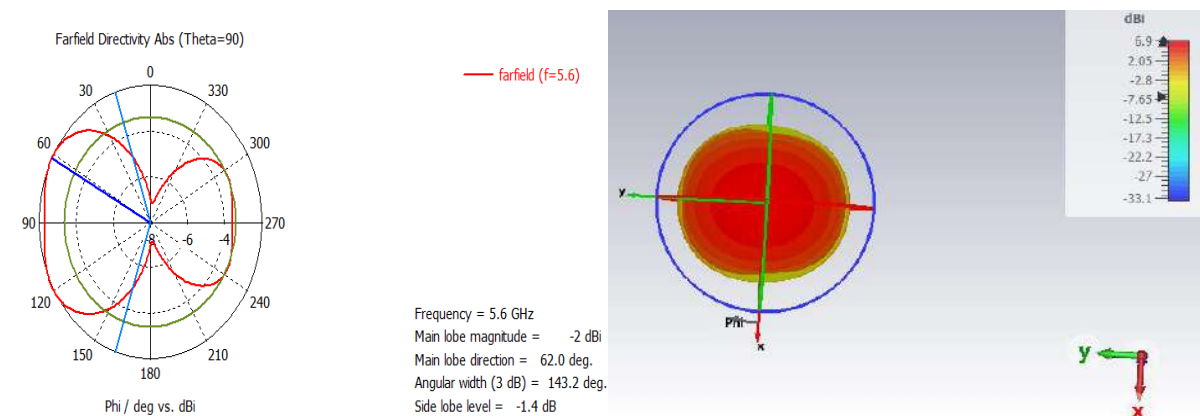


Figure 7.5.3 XY Plane [OR] $\Theta = 90^\circ$ Plane

Designed, simulated and studied the resonant frequency and the radiation pattern of the microstrip patch antenna.

EXPERIMENT NO 8

DESIGN AND SIMULATION OF PATCH ARRAY USING SINGLE FEED

Aim

To design and simulate a patch antenna array with single feed for a resonant frequency of 5.6GHz.

Design

Dielectric used = FR-4(loss free)

Dielectric constant, $\epsilon_r = 4.3$

Operating frequency, $f_r = 5.6\text{GHz}$

Height of the substrate = 1.6mm

$$\text{Width of the Patch, } W = \frac{1}{2fr\sqrt{\mu_0\epsilon_0}} \sqrt{\frac{2}{\epsilon_r + 1}} = 16.33\text{mm}$$

The effective dielectric constant of the microstrip antenna,

$$\epsilon_{\text{reff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W}\right)^{-1} = 2.43$$

Extension of the length ΔL ,

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{\text{reff}} + 0.3) \left(\frac{W}{h} + 0.264\right)}{(\epsilon_{\text{reff}} - 0.258) \left(\frac{W}{h} + 0.8\right)} = 0.7885\text{mm}$$

$$\text{The actual length of the patch, } L = \frac{1}{2fr\sqrt{\epsilon_{\text{reff}}\mu_0\epsilon_0}} - 2\Delta L = 12.33\text{mm}$$

Length (L_g) and Width (W_g) of the ground plane,

$$L_g = 6h + L = 21.93\text{mm}$$

$$W_g = 6h + L = 26.04\text{mm}$$

Let height of the ground plane, $t = 0.035\text{mm}$

Diameter of outerpin, $D = 4.7\text{mm}$

Diameter of inner pin, $d = 1.4\text{mm}$

Coaxial feed coordinates, (Xf, Yf) :

$$X_f = \frac{L}{\sqrt[2]{\epsilon_{eff}}} \quad Y_f = \frac{w}{2}$$

Procedure

- i. Open CST Microwave Studio → New Template → Select Microwaves & RF/Optical as the application area → Select Antennas → Click Next → Select Planar workflow → Click Next → Select the Time domain Solver for simulation → Click Next → Select the units → Click Next → Set and define the frequencies → Click Next → Click Finish → The template is now ready.
- ii. Calculate the radius of the coaxial feed using 'Calculate analytical Line impedance' in the 'Calculate' option inside the 'Macros' tab.
- iii. Draw the required antenna structure using 'Modelling tool'.
- iv. Assign the port using 'Pick Face' option, then click 'Apply' in the 'Waveguide Port' tab
- v. Set the field monitors → In that setup, set the farfield monitor at 5.6GHz.
- vi. Start the simulation.
- vii. Analyse the reflection coefficient(S11) and the step impedance. Change values of (Xf,Yf) for desired S11
- viii. Analyse the 3-Dimensional radiation pattern and the farfield cuts and check the pattern and cuts
- ix. Cross verify the 3-Dimensional results with the 2-Dimensional results in the 3 cut planes($\Phi = 0^\circ$, $\Phi = 90^\circ$, $\theta = 90^\circ$)

Observation

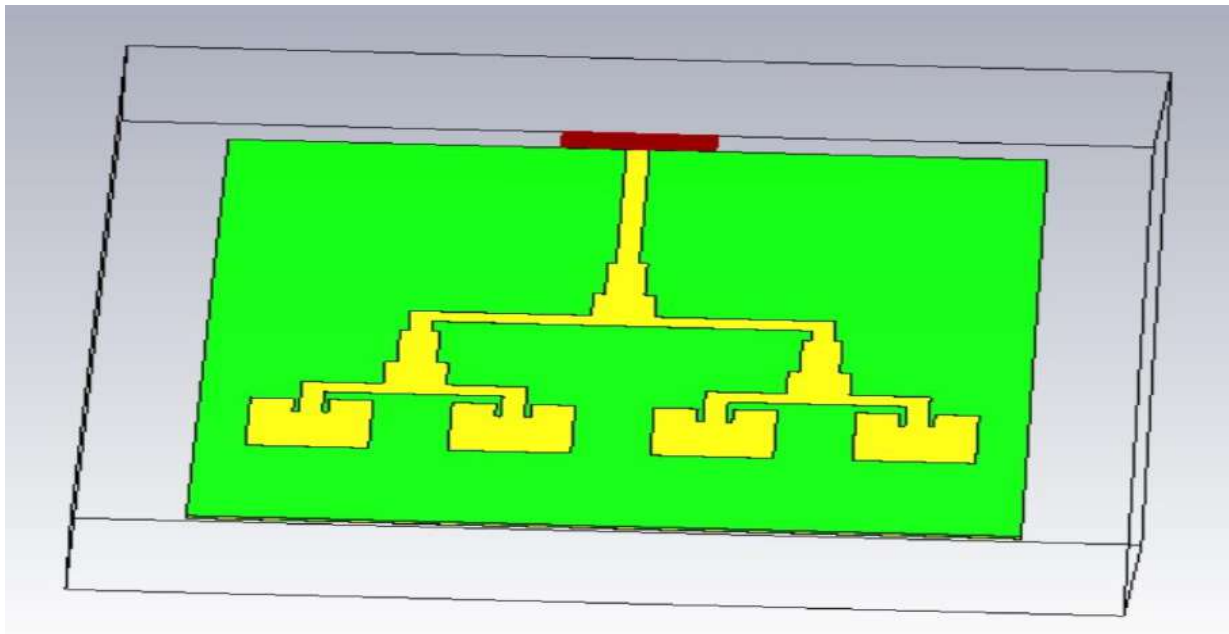


Figure 8.1 Schematic of antenna array using single feed

Results

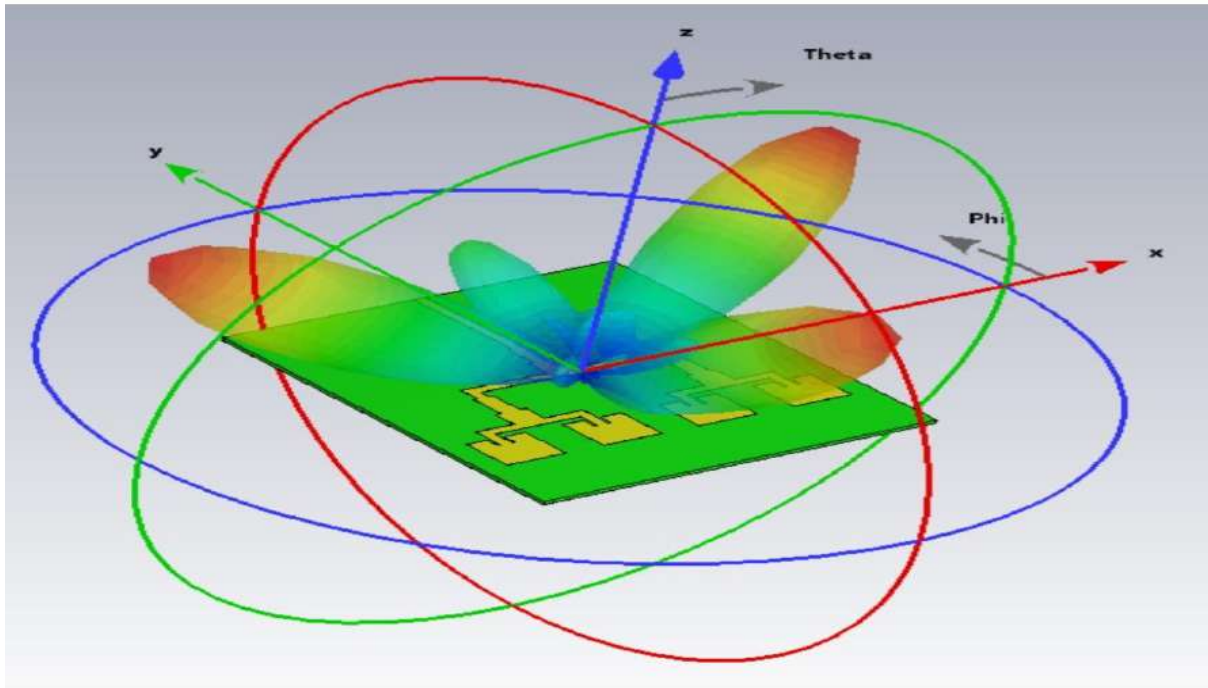


Figure 8.2 3-Dimensional radiation pattern

Far field cuts

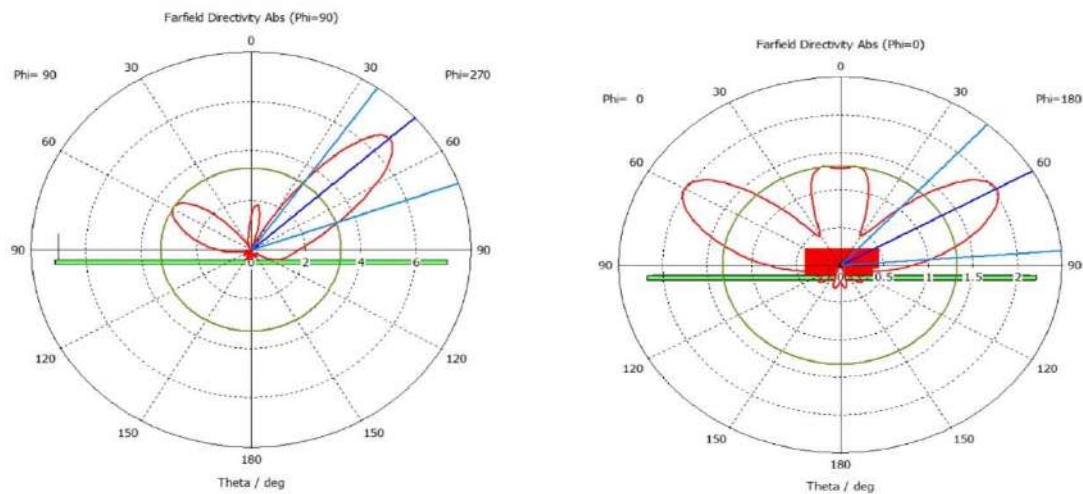


Figure 8.3 Far field cuts of antenna

Patch array using single feed is designed and simulated. Radiation pattern is analyzed.